

Passive cardiac shear-wave elastography, part 3: all 15 labelled windows, and M-lines redrawn on the focused-beam buffer

shearWaveProcessing – study/analysis/passive_atlas_all15.py, draw_labelled_mlines_b3.py, mline_difference_check.py, buffer_registration.py

24 September 2026

Summary

- **All 15 hand-labelled windows** are now used, grouped by their velocity-panel confidence score (treated as blind): 6 *clear*, 7 *plausible*, 2 *guess/none*. Parts 1 and 2 used 7 of them.
- **M-lines were redrawn on buffer 3** (focused beams, REFoCUS) on the frame at the event’s cardiac phase.
- **The new lines are not better.** The buffer-3 frames come from a heartbeat 2–3 beats (1.7–3.2 s) before the buffer-4 event. The heart has moved by a median of **3.1 mm** relative to buffer 4 at the event, against 0.8 mm for the buffer-1 frames (1–2 beats earlier).
- **Ruled out as causes:** the reconstructions themselves (phantom: buffers agree to $< 35\ \mu\text{m}$) and the phase offset of the chosen frame (0.2 mm).
- **In the data that is processed**, the old lines lie slightly more in coherent tissue. Several windows lose their wavefront on the new line (C000000002, 039, 004).
- **The recipe conclusions of parts 1–2 hold on all 15 windows and both line sets:** velocity; low band corner at 15 Hz (15–90/100 Hz separates best); light Gaussian; no directional filter; no SVD beyond one IQ component; no CFWI.

1 The redrawn M-lines, and why they differ

Timing: every drawing frame is from another heartbeat

Only buffers 4, 2(+5) and 6 are R-peak triggered. The trigger log places the frames drawn on, relative to the buffer-4 event:

- buffer 1: 0.8–2.2 s before, 1–2 beats earlier;
- buffer 3: 1.7–3.2 s before, 2–3 beats earlier.

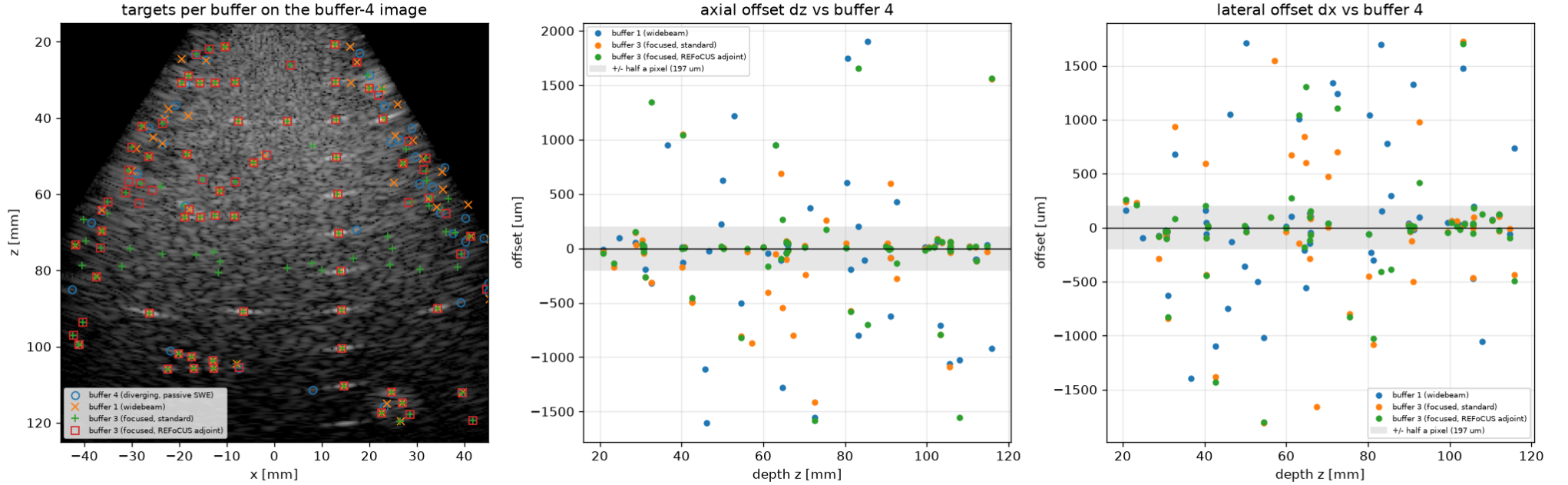
The nearest-phase buffer-3 frame is 0–20 ms off the event phase: buffer 3 has only 26–32 frames per acquisition. For buffer 1 the offset is 5 ms or less.

Reconstruction offsets: ruled out on the resolution phantom

In a static phantom there is no motion, so any displacement of a point between buffers is a reconstruction or geometry offset. Wire targets were located with sub-pixel precision in each buffer and matched to buffer 4 (21–25 wires, all ≥ 20 dB above background):

buffer vs buffer 4	dx median [μm]	dz median [μm]	dz IQR [μm]	depth scaling	lateral scaling
buffer 1 (widebeam)	7	8	103	0.07 %	0.70 %
buffer 3 (focused, standard)	-34	10	45	0.21 %	0.21 %
buffer 3 (focused, REFoCUS)	6	5	13	0.05 %	0.42 %

All offsets are a tenth of a pixel ($394\mu\text{m}$) or less. At most 0.15 mm at 20 mm off-axis comes from scaling. **The buffers map the same point to the same place.**



In vivo: what moves, and by how much

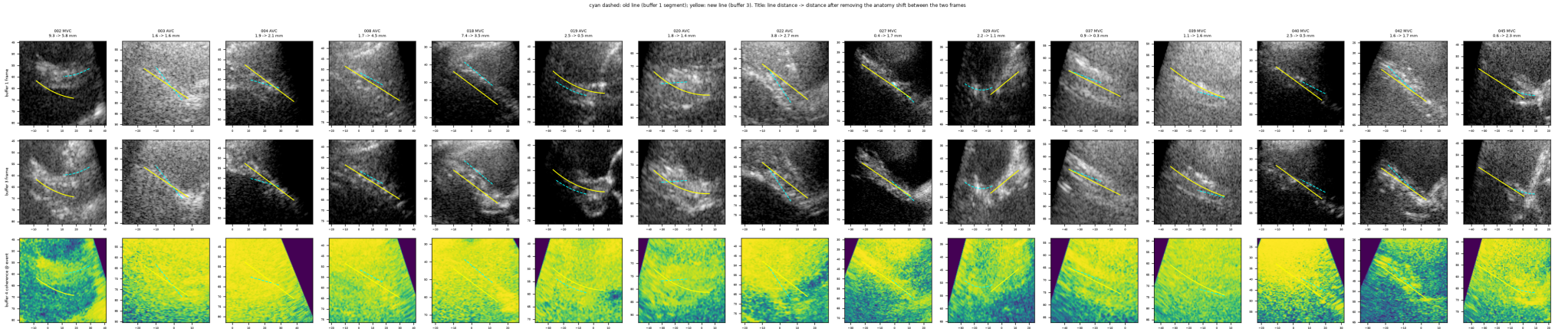
For each window, the anatomy was registered between the frames by phase correlation. The images were band-passed to 1–6 mm structure, and a known-shift self-test was run on every window.

quantity (median over 15 windows)	value	meaning
phase offset alone (buffer 4 at event vs event + offset, same beat)	0.2 mm	negligible
buffer-1 frame vs buffer 4 at the event	0.8 mm	1–2 beats apart
buffer-3 frame vs buffer 4 at the event	3.1 mm	2–3 beats apart; larger than buffer 1 in 12/15
new vs old line, where they overlap	1.8 mm	only ~50 % overlap; angle 1–28°
after removing the anatomy shift between the two frames	1.7 mm	the remaining part is interpretation
buffer-4 coherence along the line (tissue ≈ 1)	0.90 old / 0.88 new	new higher in 3/15

Reading. The difference is partly beat-to-beat motion and partly interpretation. Where the buffer-3 anatomy moved most, removing the shift brings the lines together:

- C000000018: 7.4 $\rightarrow$ 3.5 mm;
- C000000019: 2.5 $\rightarrow$ 0.5 mm;
- C000000040: 2.5 $\rightarrow$ 0.5 mm.

Elsewhere the new lines cover a different stretch or run at a different angle. The in-vivo registrations are cross-modality (correlation 0.48–0.86), so single large values are approximate; C000000020's 11.7 mm (correlation 0.56) is the least certain.

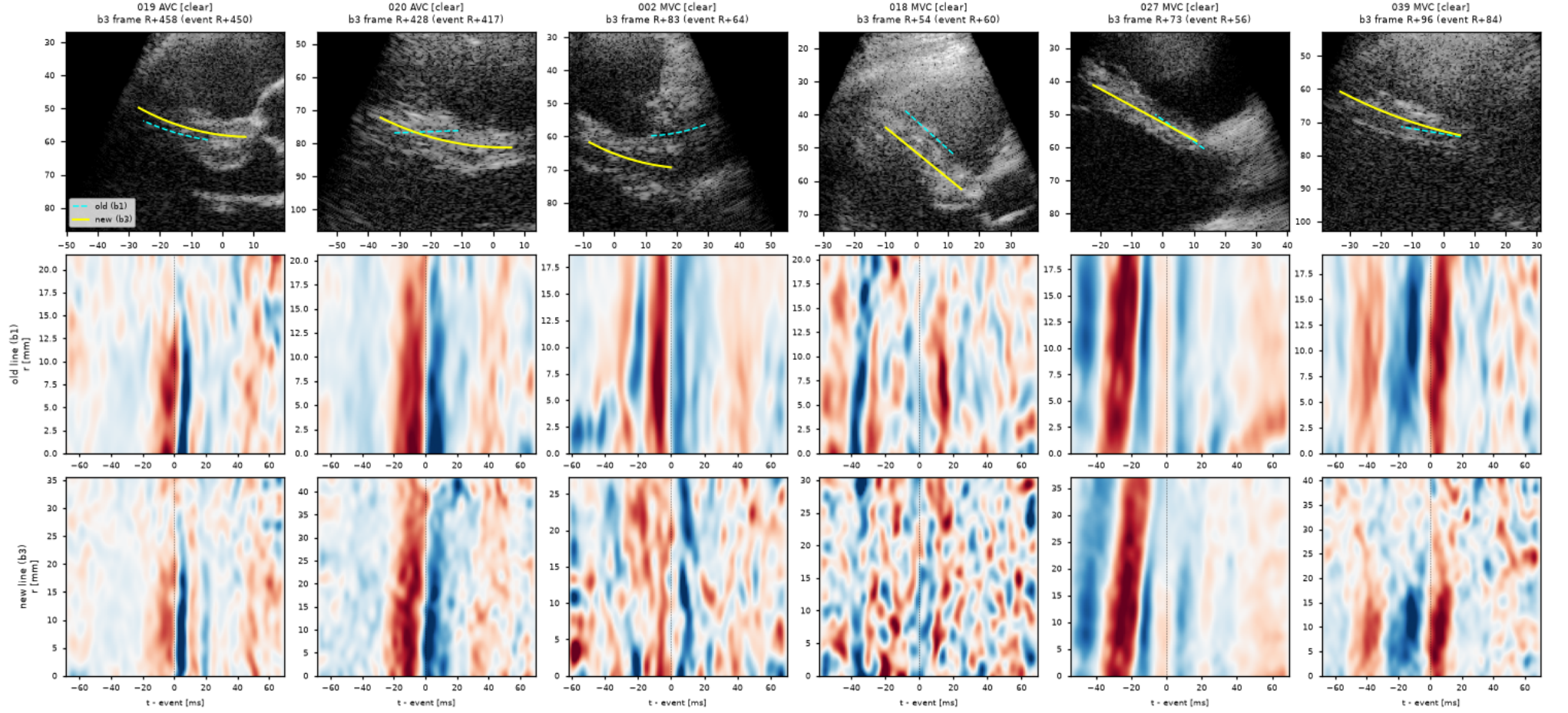


2 Old versus new M-line: the space-time plots

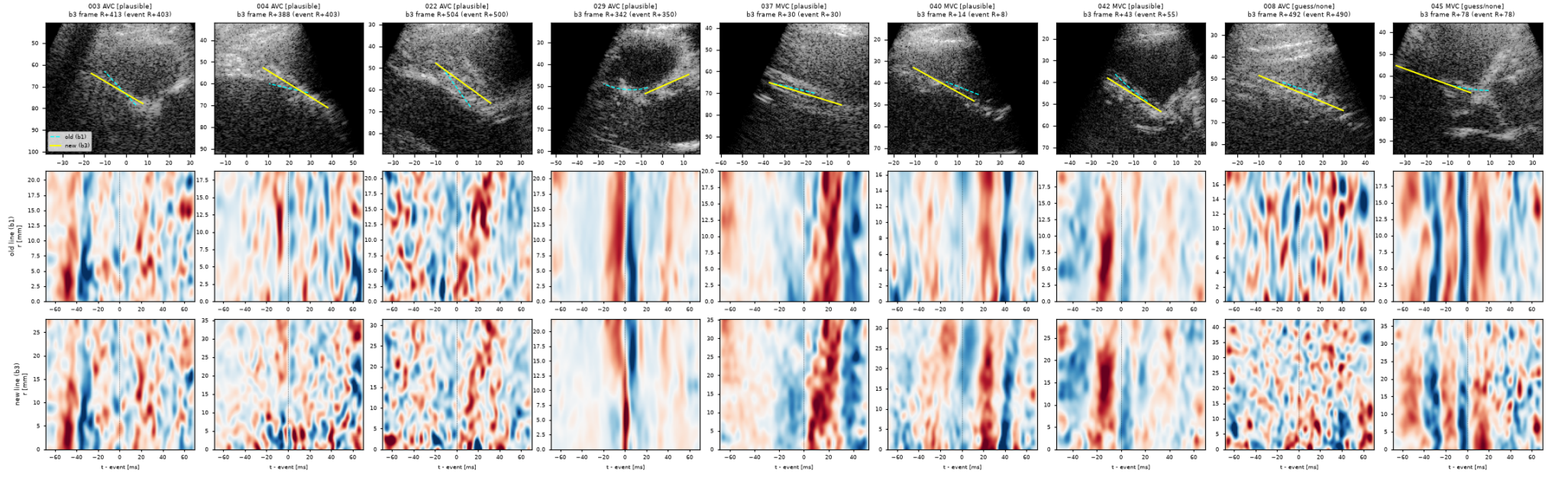
Default recipe (velocity 15–150 Hz, Gaussian 0.6×1.2 mm, mean 3, 5 lines) on the old buffer-1 segment and on the new buffer-3 line. Top row: the buffer-3 frame with both lines.

- Most windows look alike on both lines (019, 020, 027, 029, 037, 040, 042).
- Some lose their front on the new line: C000000002 (whose buffer-3 anatomy is 4.8 mm off), 039 and 004.
- None becomes clearly better.
- The new lines are longer (27–43 mm vs 16–22 mm), which helps a slope reading only where the line stays in the wall.

Old (buffer-1 segment) vs new (buffer-3) M-line, default velocity 15-150 Hz recipe



Old (buffer-1 segment) vs new (buffer-3) M-line, default velocity 15-150 Hz recipe



3 Recipe scores on all 15 windows, both line sets

Per panel: the tracking score of the strongest straight line (mean $|\text{signal}|$ on it / panel RMS). No hand line is needed, so it works on both line sets. Medians per group; the separation is clear minus guess/none.

- A recipe that separates waves from noise has a high separation.
- A recipe that invents structure raises the guess/none group too.
- The guess/none group has only two windows, so small differences are noise.

recipe	old lines (buffer 1)				new lines (buffer 3)			
	clear	plaus.	g/n	sep.	clear	plaus.	g/n	sep.
v1 displacement 10–150 (old view A)	2.41	1.97	1.80	0.61	2.09	1.88	1.60	0.49
v1 velocity 10–150	2.39	2.19	1.90	0.49	2.17	2.01	1.64	0.53
v1 acceleration	2.19	1.71	1.95	0.24	2.35	1.58	1.06	1.29
v1 displacement 15–100	2.57	2.33	1.86	0.70	2.33	2.00	1.58	0.75
v1 displacement 5–150	1.95	1.70	1.63	0.32	1.80	1.53	1.62	0.17
v1 velocity 15–100	2.47	2.25	1.87	0.61	2.28	2.15	1.64	0.64
v1 directional, displacement toward low r	2.27	1.72	1.68	0.59	1.91	1.79	1.68	0.23
v1 polynomial detrend + low-pass	2.19	1.99	2.04	0.15	1.86	1.81	1.72	0.14
v1 view C (velocity 15–90, Gauss 1.0, mean 5, 9 lines)	2.57	2.49	1.88	0.69	2.47	2.23	1.76	0.72
v1 Petrescu/Santos acceleration	2.74	1.87	1.67	1.08	2.31	1.79	1.53	0.79
v2 default (velocity 15–150, Gauss 0.6, mean 3, 5 lines)	2.40	2.24	1.90	0.50	2.25	2.17	1.60	0.64
v2 Gaussian 1.0×2.0	2.56	2.37	2.15	0.42	2.41	2.21	1.76	0.66
v2 Gaussian 2.0×4.0	2.65	2.57	2.17	0.48	2.45	2.35	1.82	0.63
v2 moving mean 9	2.43	2.23	1.65	0.78	2.22	2.07	1.64	0.59
v2 9 lines $\times$ 0.8 mm	2.47	2.31	1.91	0.56	2.44	2.16	1.64	0.80
v2 combined medium	2.58	2.43	2.06	0.53	2.43	2.20	1.77	0.66
v2 IQ SVD, 1 component	2.39	2.29	1.81	0.58	2.30	2.10	1.75	0.55
v2 IQ SVD, 3 components	2.44	2.37	2.13	0.30	2.17	2.04	1.77	0.41
v2 IQ SVD instead of the high-pass	1.77	1.66	1.83	−0.05	1.57	1.65	1.79	−0.22
v2 CFWI 2 cm/s	2.33	2.06	1.83	0.50	1.86	1.97	1.87	−0.01

Reading.

- **Consistent on both line sets:**
 - a 15 Hz low corner separates better than 5–10 Hz;
 - view C is the best production view;
 - the 5 Hz high-pass, polynomial detrending, SVD as the motion filter and CFWI separate poorly;
 - more than one SVD component lowers the separation.
- **Acceleration** separates well by this score, because noise panels offer no straight line. Visually it is still the noisiest quantity (part 1).
- **The new lines** lower the clear-window scores for most recipes. They also lower the guess/none scores, so the separation is similar – no gain in wave visibility.
- The full table (all 83 recipes $\times$ 2 line sets) is in `study/logs/passive_atlas15_scores.csv`.

4 Conclusions

1. **Keep the buffer-1 lines for now**, or motion-correct the buffer-3 lines: shift each by its measured buffer-3 $\rightarrow$ buffer-4 anatomy offset. The focused image is sharper, but it is 2–3 beats away from the data being processed, and that costs more than the sharper image gains.
2. **For future acquisitions**, acquire the focused buffer directly before or after buffer 4, in the same or the adjacent beat, so the M-line image and the passive data share the heart position.
3. **The default-pipeline candidate from part 2 stands on all 15 windows:**
 - velocity, band-pass from 15 Hz (15–90 to 15–150 Hz);
 - Gaussian 0.6–1.0 mm;
 - moving mean 3–5;
 - 5–9 M-lines;
 - no directional filter, no SVD, no CFWI.
4. **Caveats.**
 - The scores are from one observer.
 - The guess/none group has two windows.
 - The best-line score is a proxy that rewards any straight structure, which is why the plots, not the numbers, carry the decision.

Reproduce.

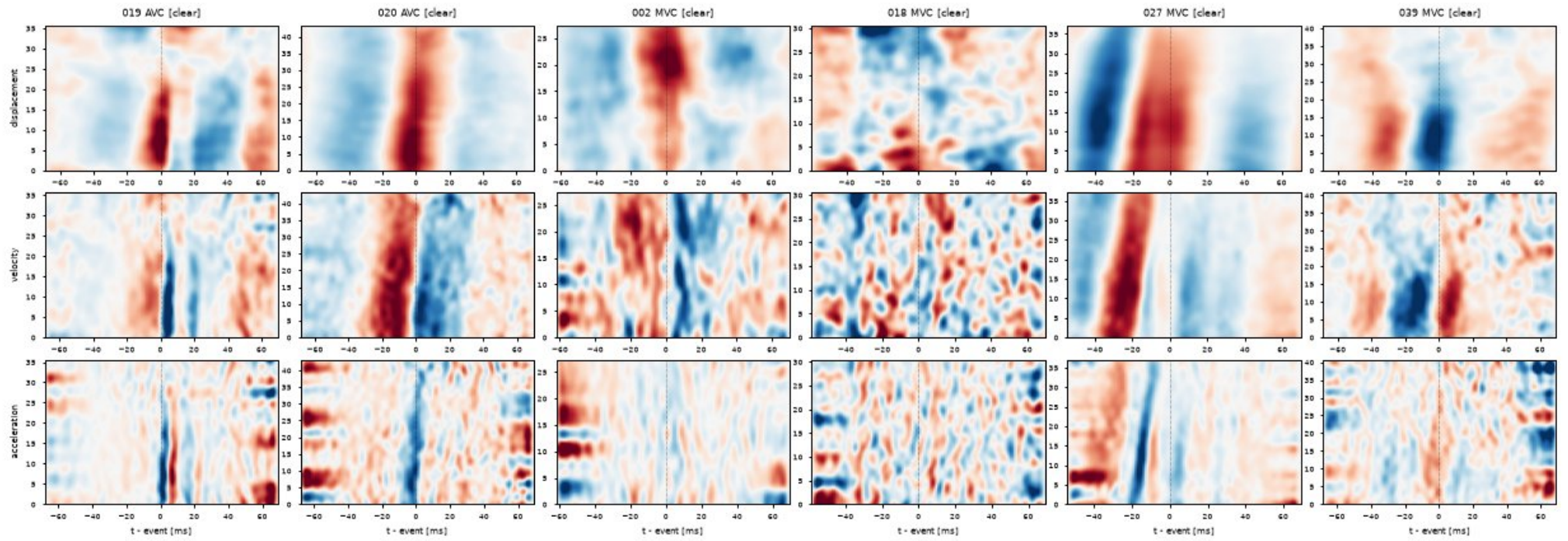
- `draw_labelled_mlines_b3.py` (interactive);
- `buffer_registration_phantom.py`;
- `mline_difference_check.py`;
- `passive_atlas_all15.py` (~ 70 min; `--figures` re-renders from the cache);
- `tectonic_passive_methods_v3.tex`.

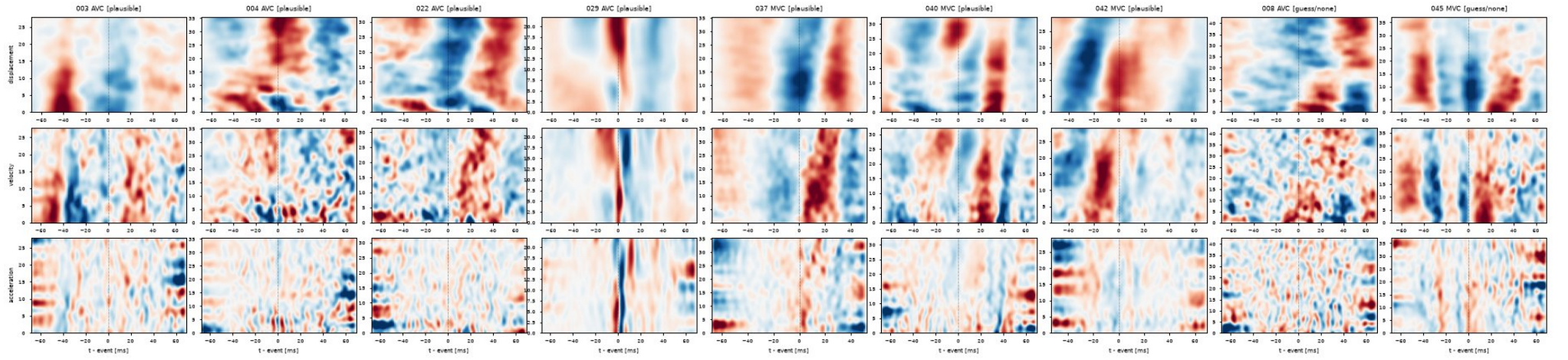
A All recipe families on the new (buffer-3) lines

Each family is shown twice: the six *clear* windows, then the seven *plausible* and two *guess/none* windows. The recipes and their meaning are as in parts 1 (v1: view-A displacement base) and 2 (v2: velocity 15–150 Hz base).

v1 – quantity: displacement, velocity, acceleration

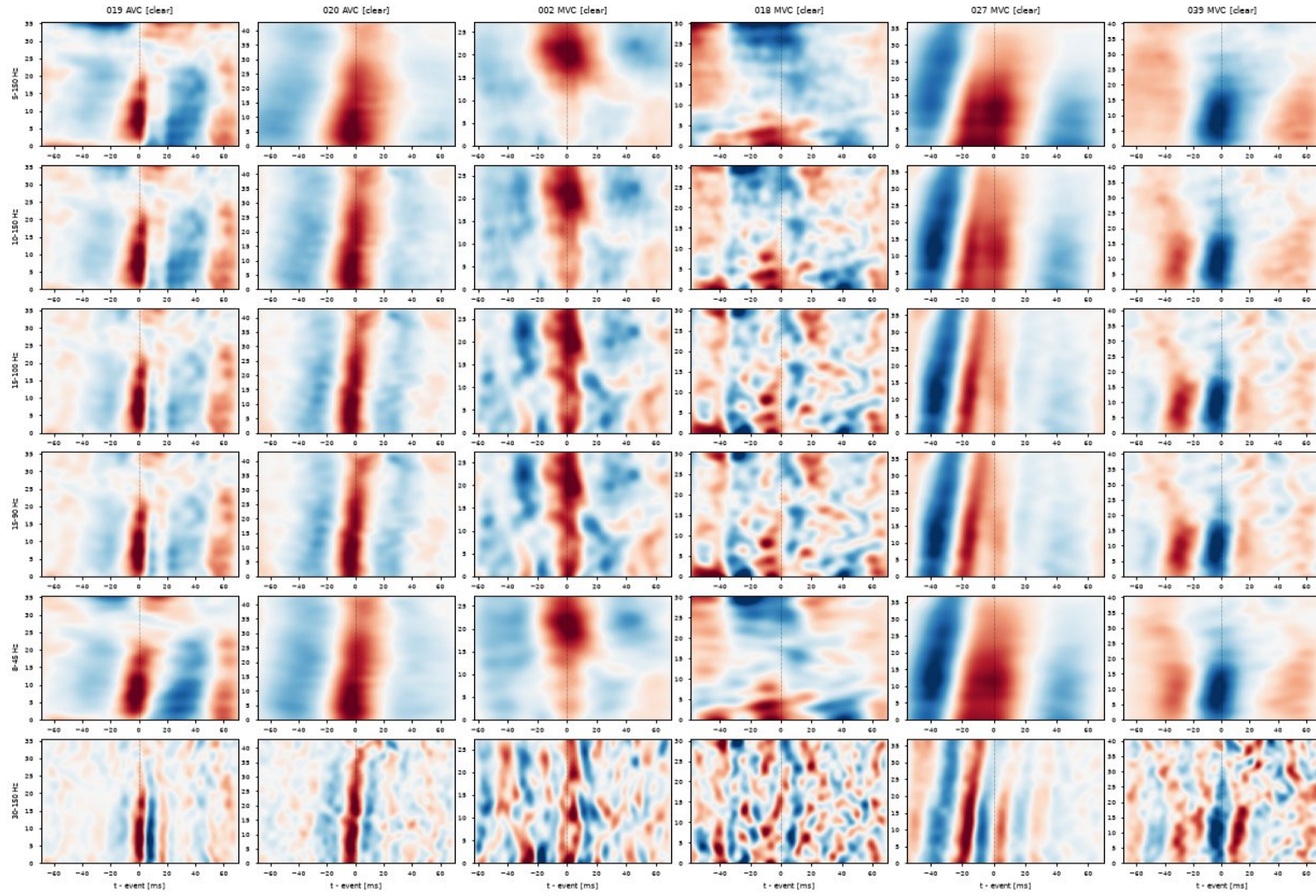
v1 / quantity - clear windows - b3 M-lines

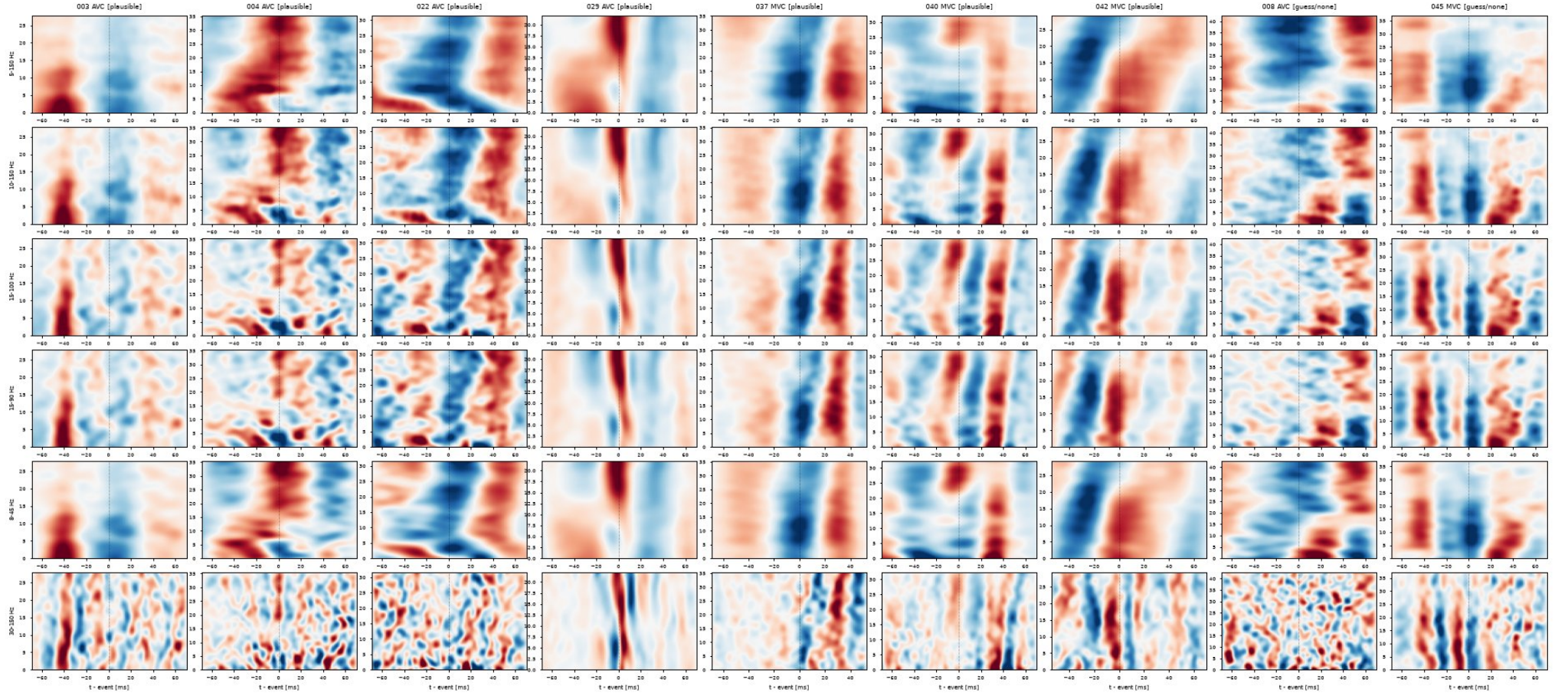




v1 – band-pass, displacement

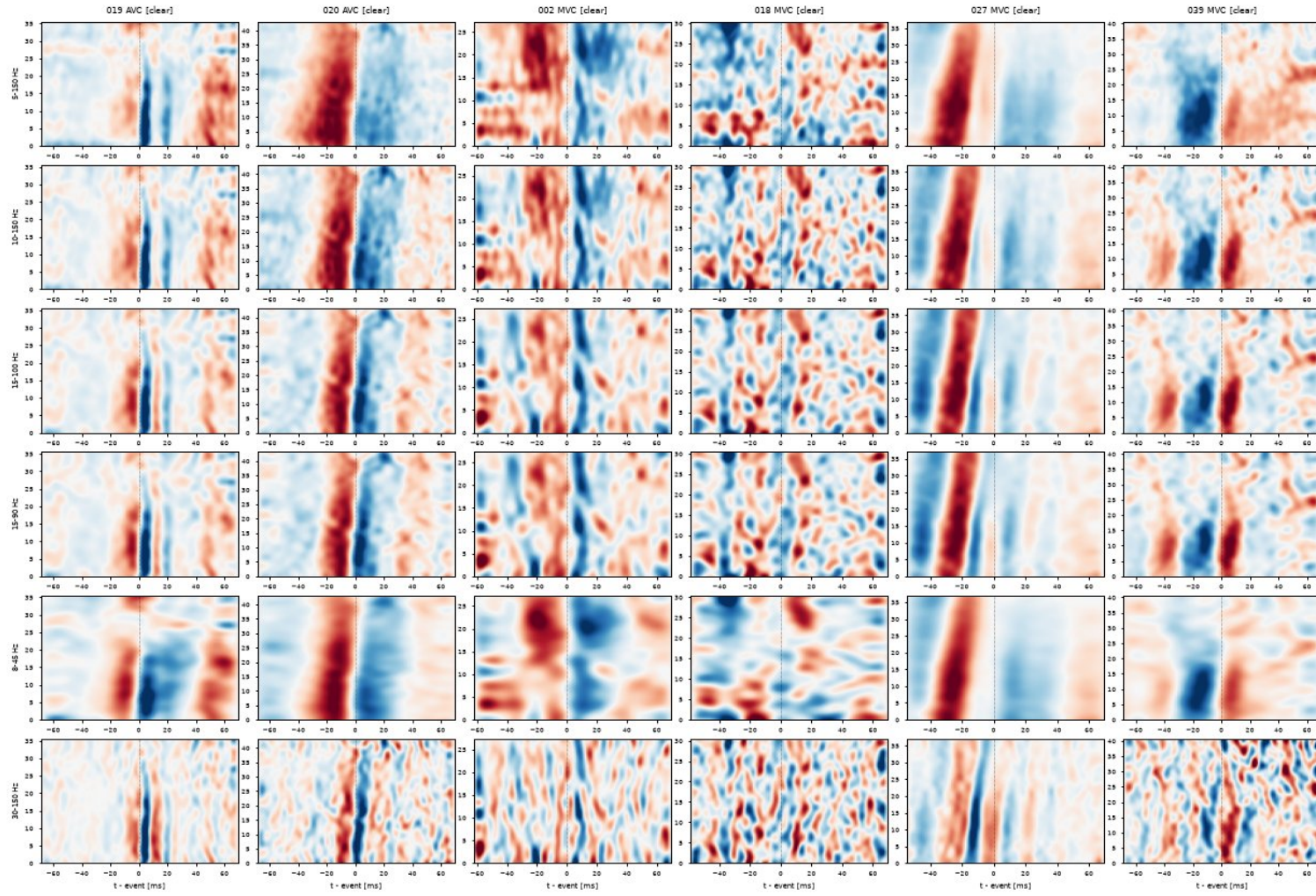
v1 / band_displacement - clear windows - b3 M-lines



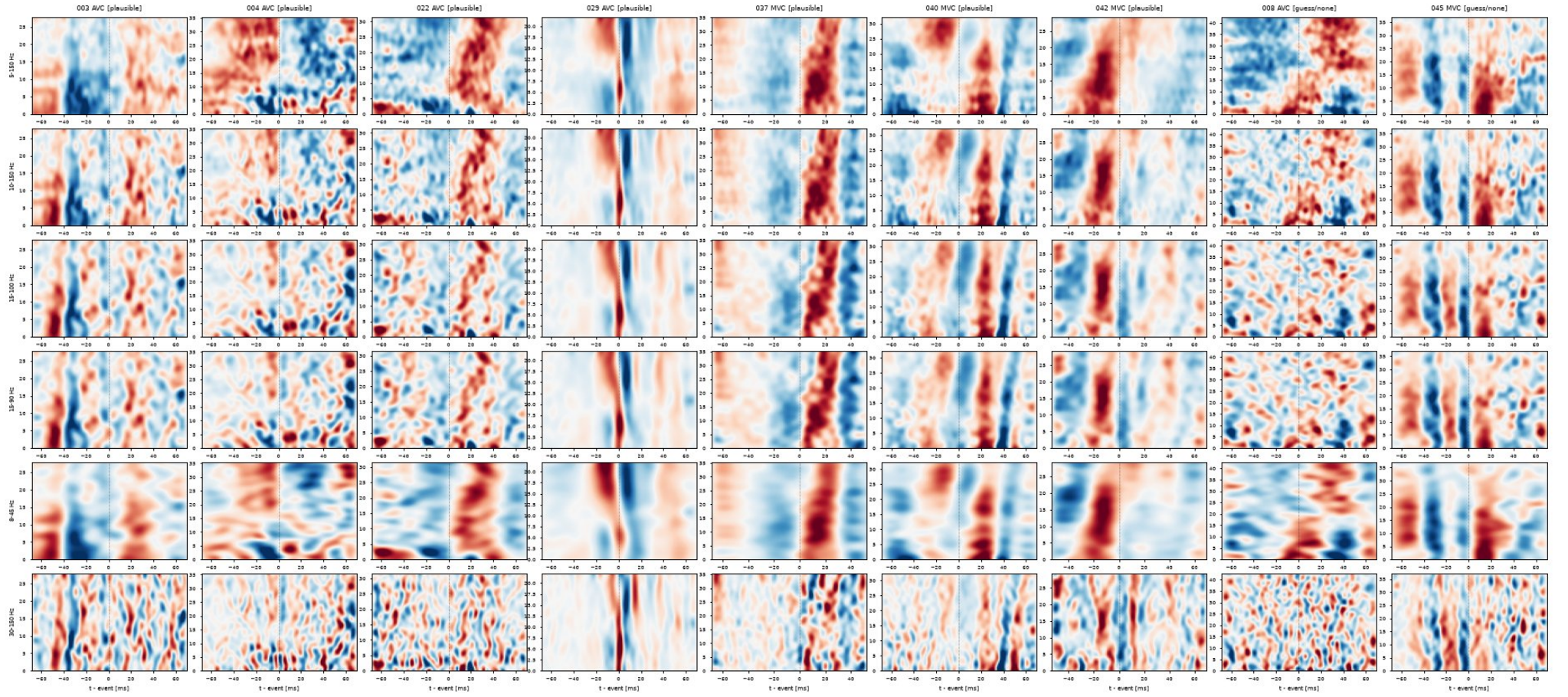


v1 – band-pass, velocity

v1 / band_velocity - clear windows - b3 M-lines

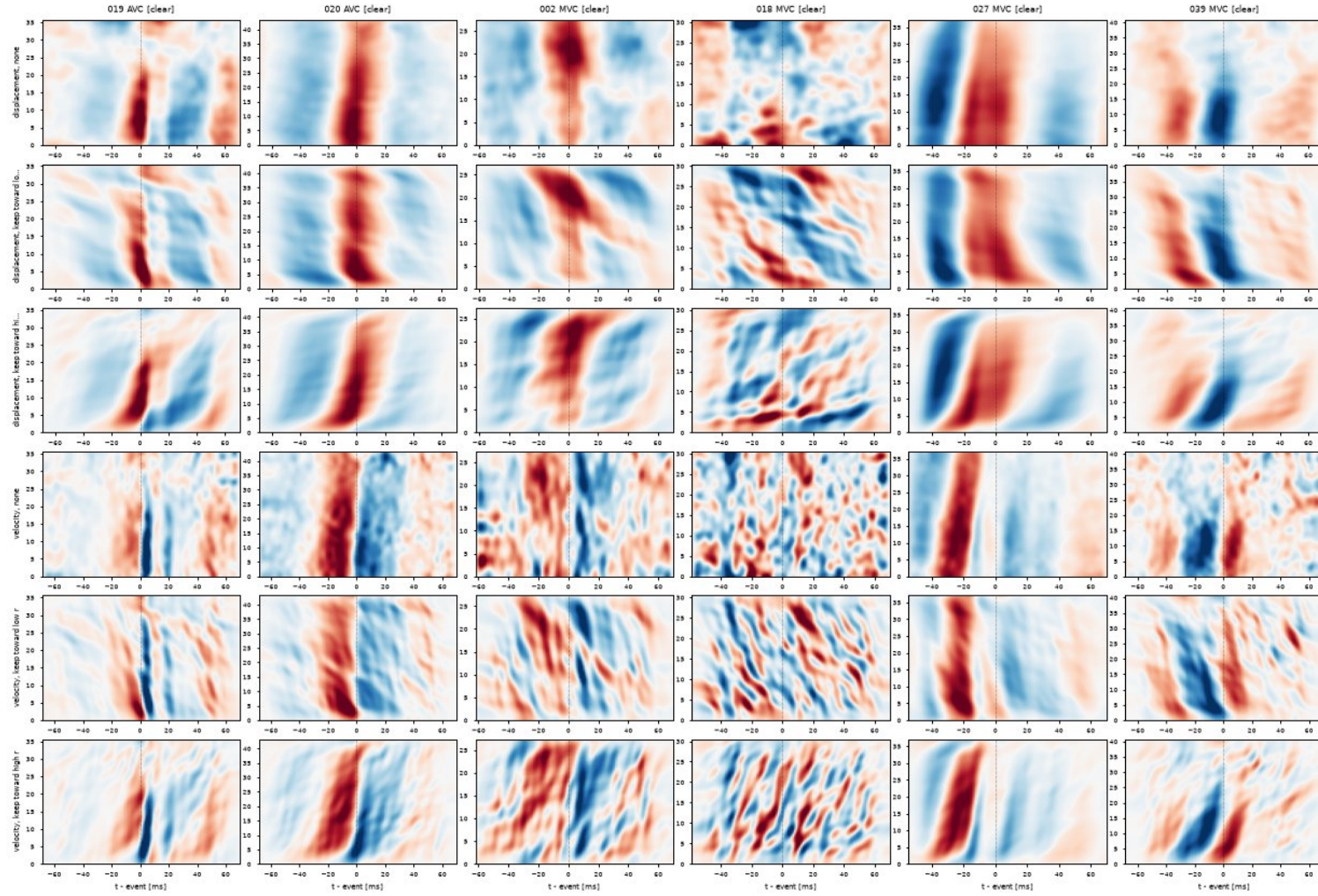


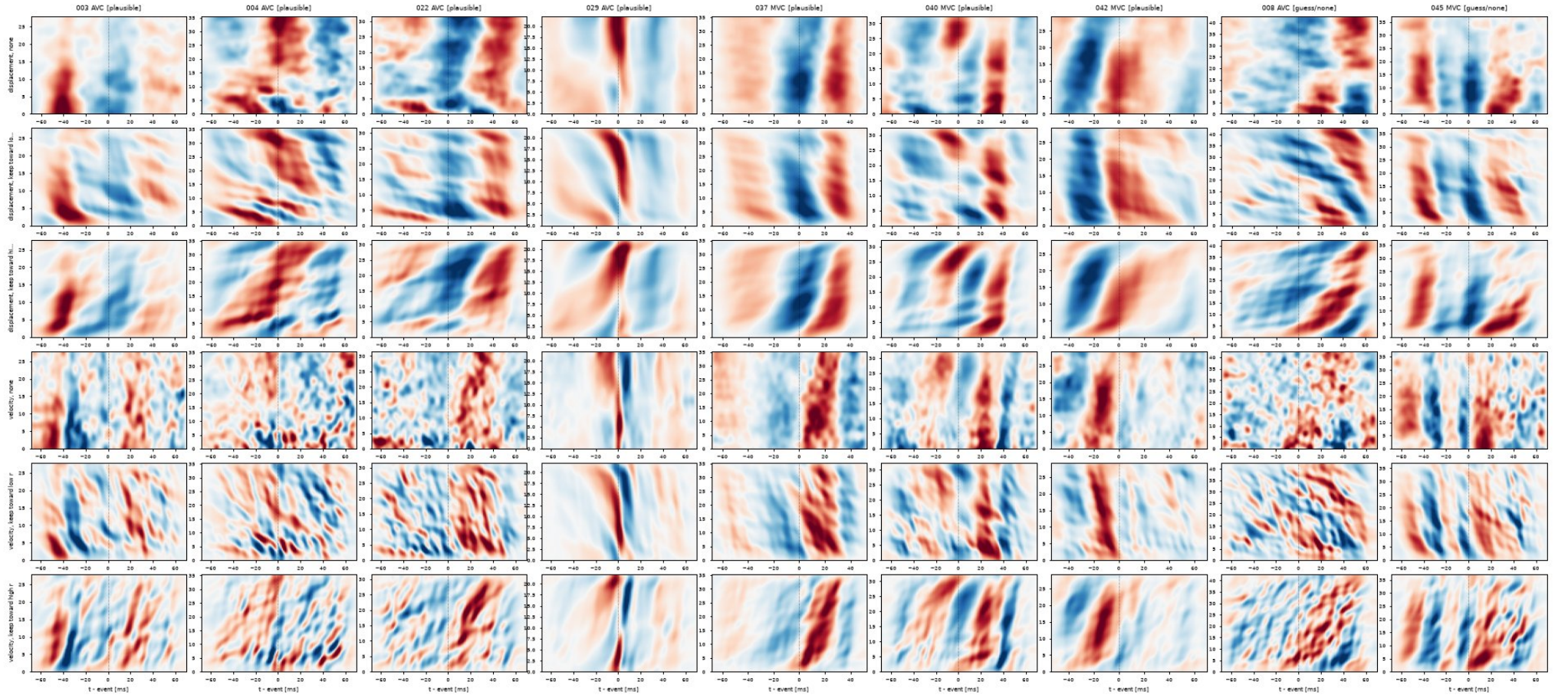
v1 / band_velocity - plausible + guess/none windows - b3 M-lines



v1 – directional filter

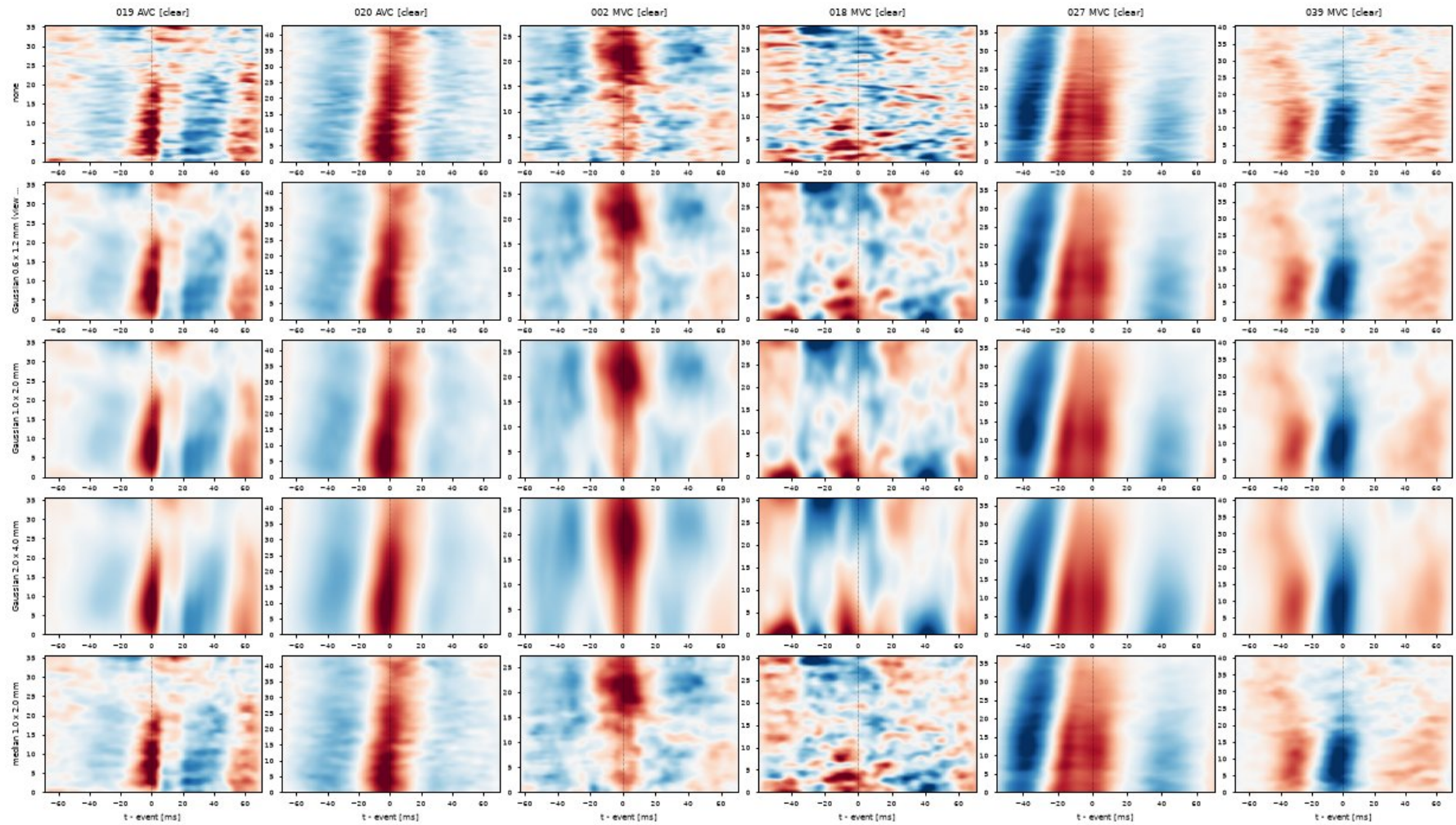
v1 / directional - clear windows - b3 M-lines

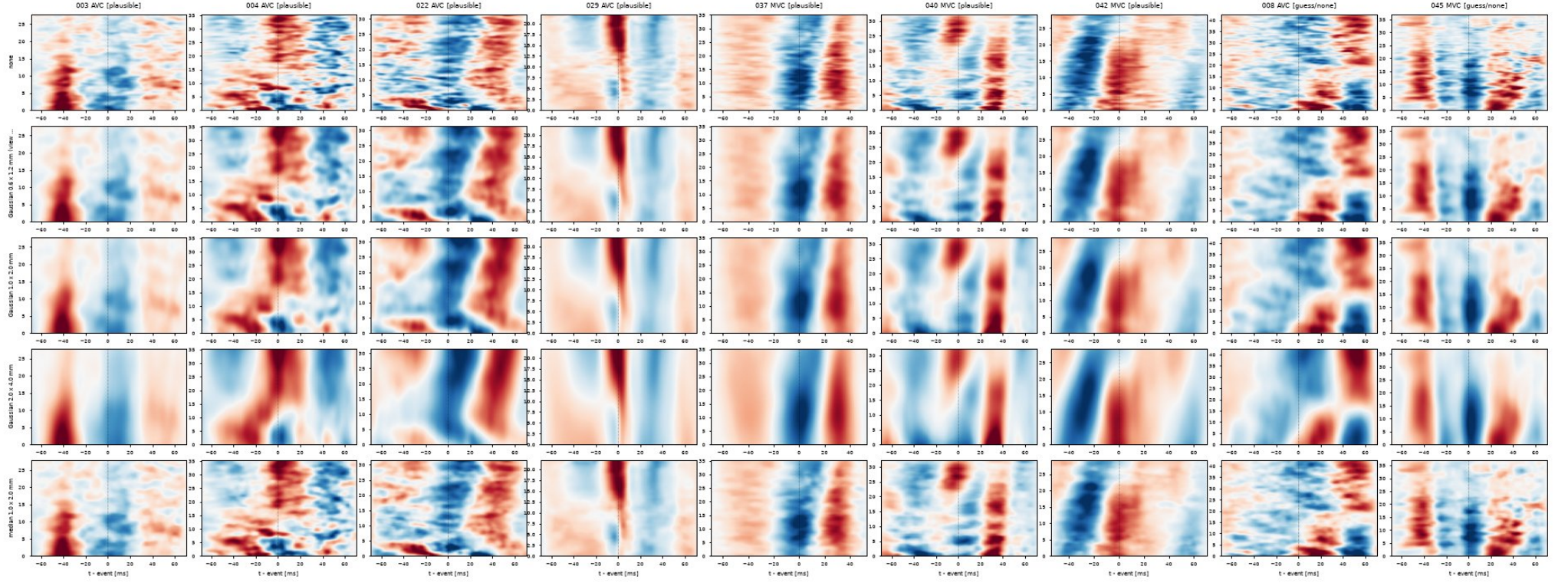




v1 – spatial smoothing (displacement)

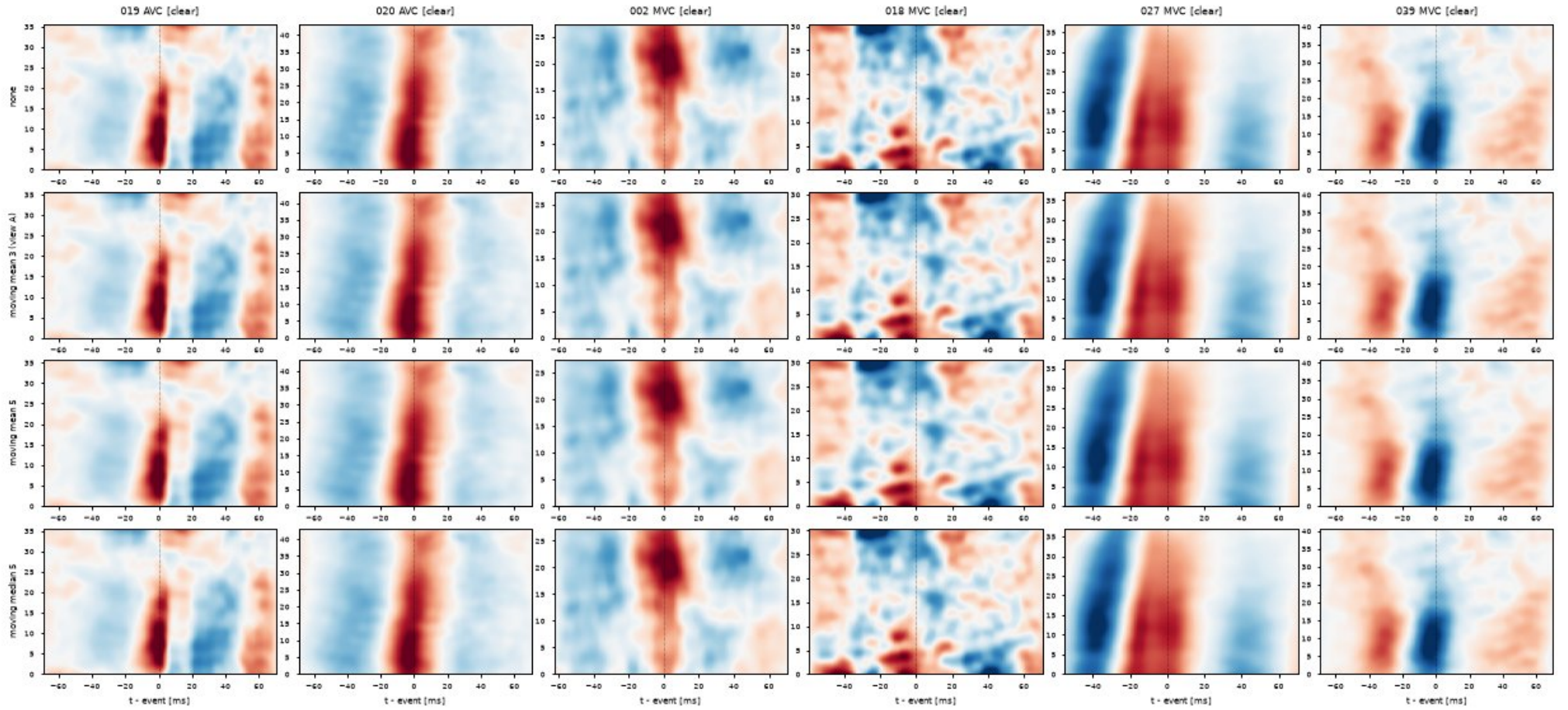
v1 / spatial - clear windows - b3 M-lines

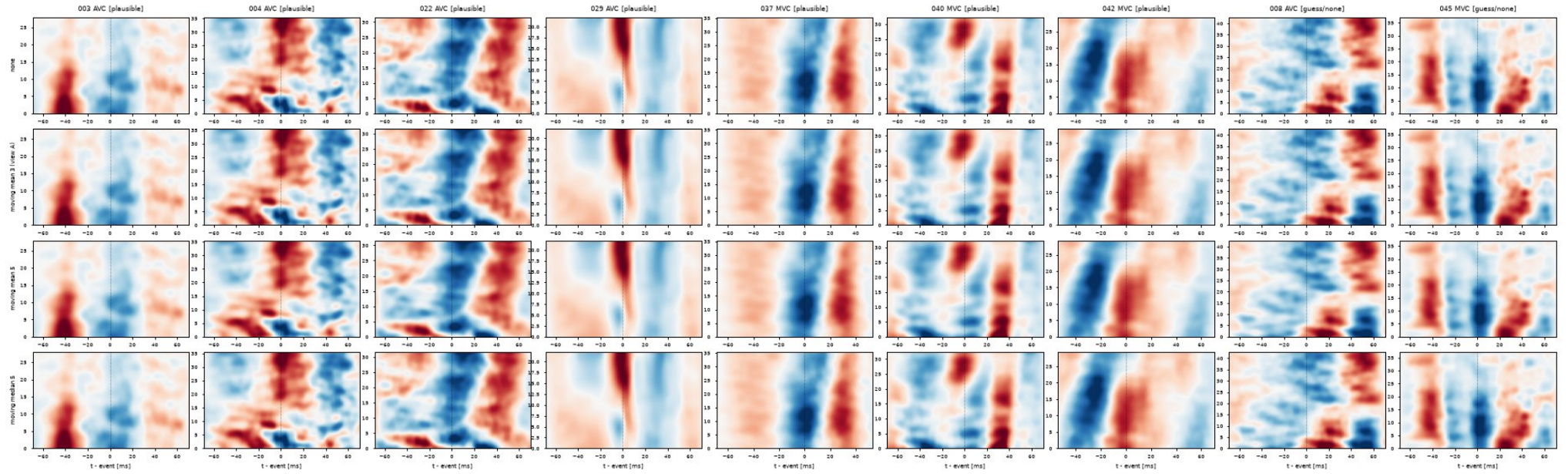




v1 – temporal smoothing (displacement)

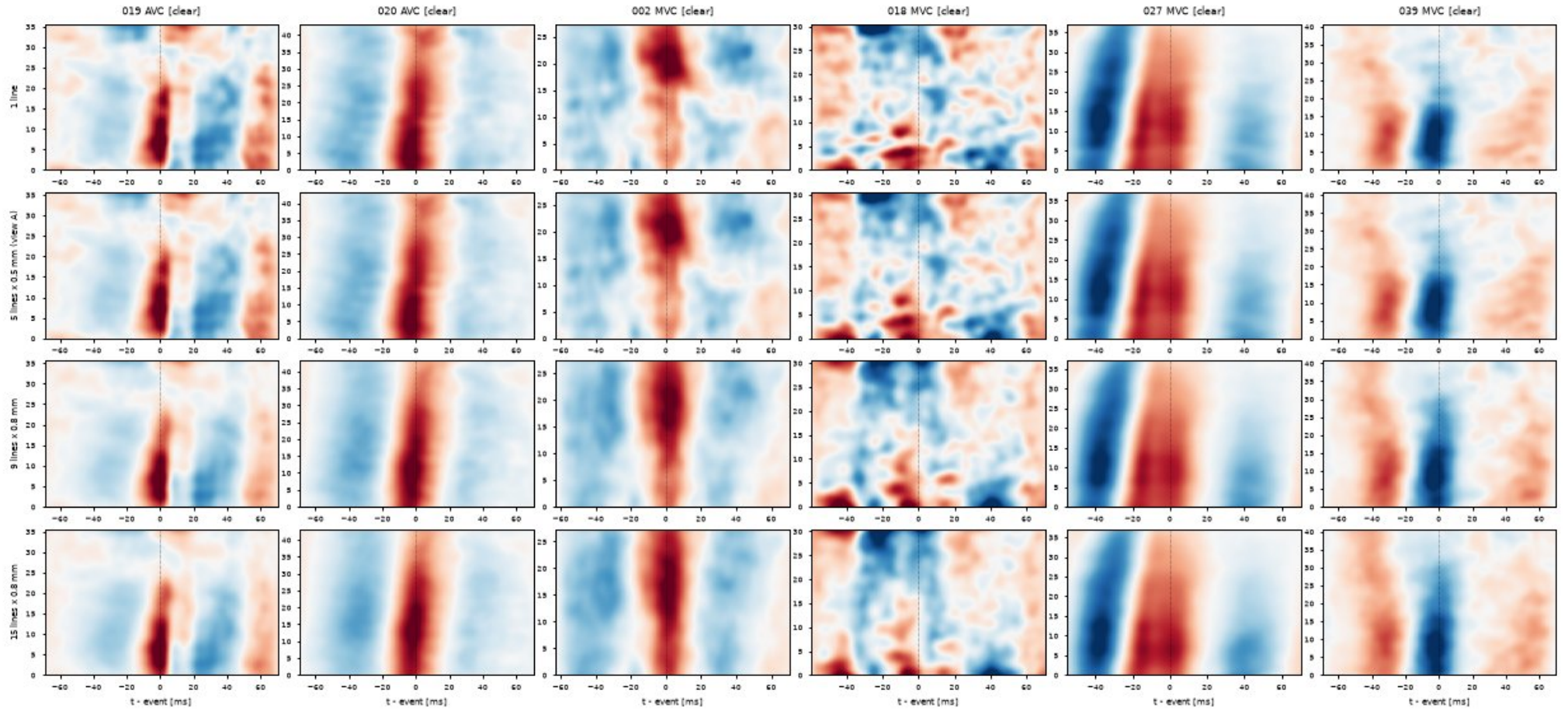
v1 / temporal - clear windows - b3 M-lines

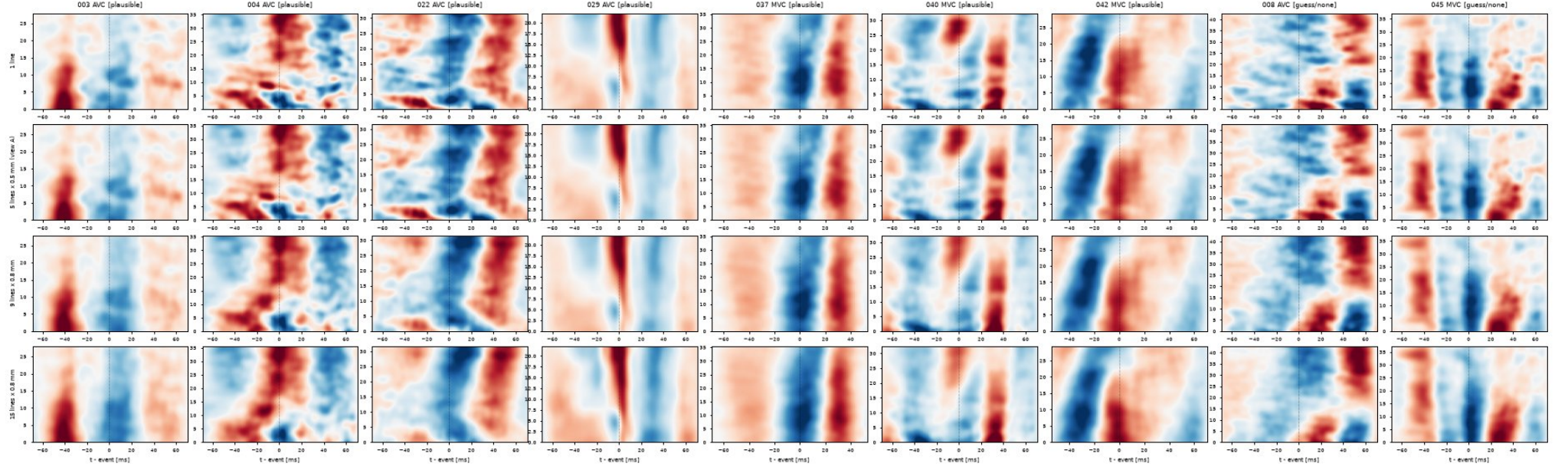




v1 – M-line averaging (displacement)

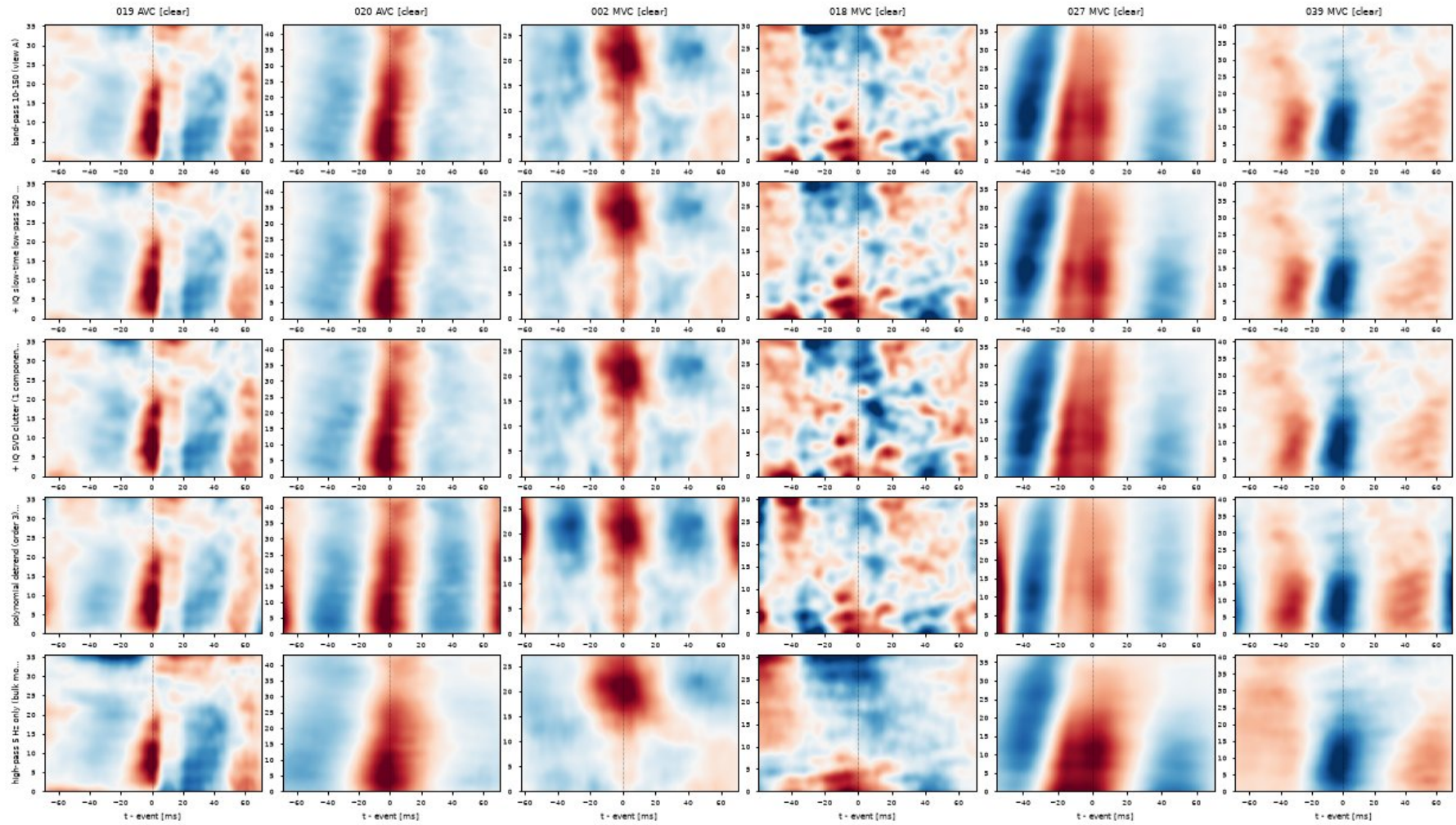
v1 / mline - clear windows - b3 M-lines

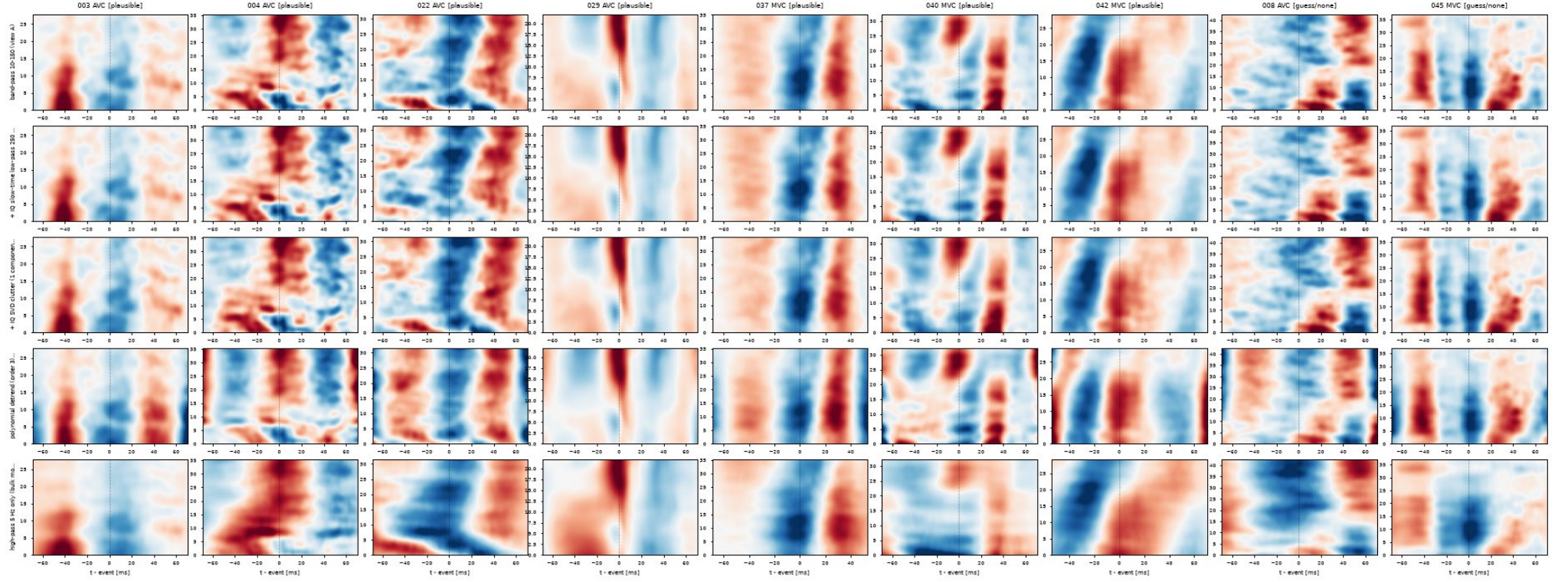




v1 – motion and clutter handling

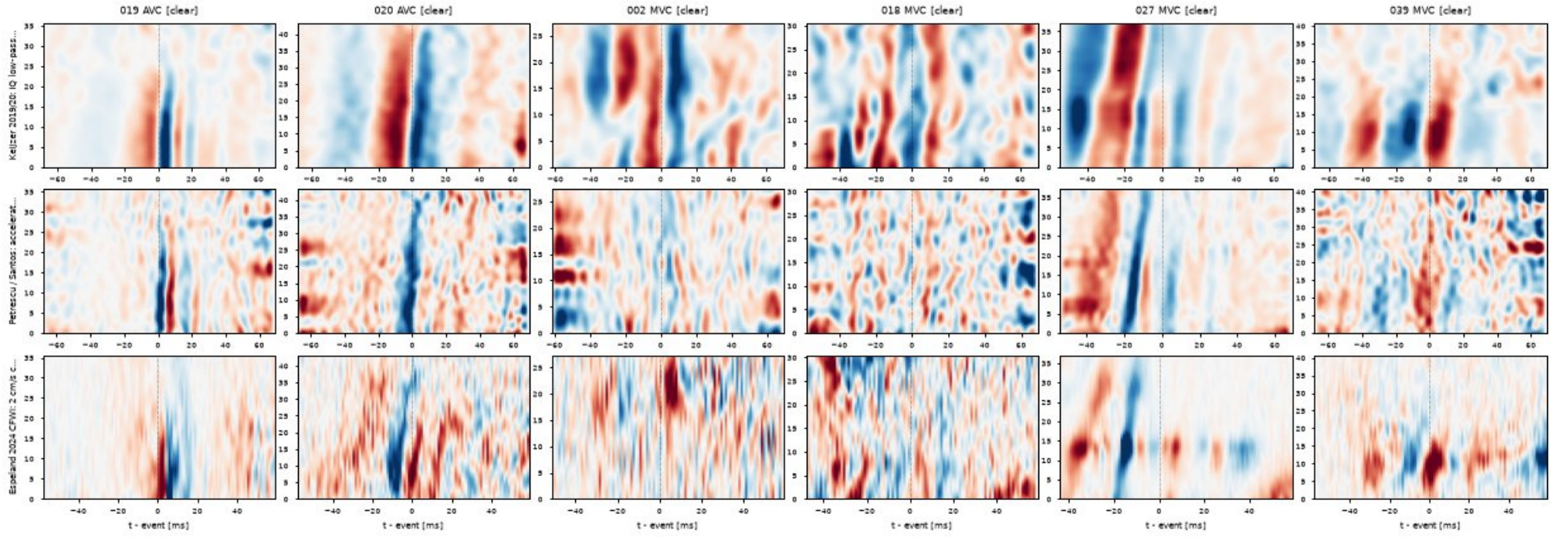
v1 / motion - clear windows - b3 M-lines

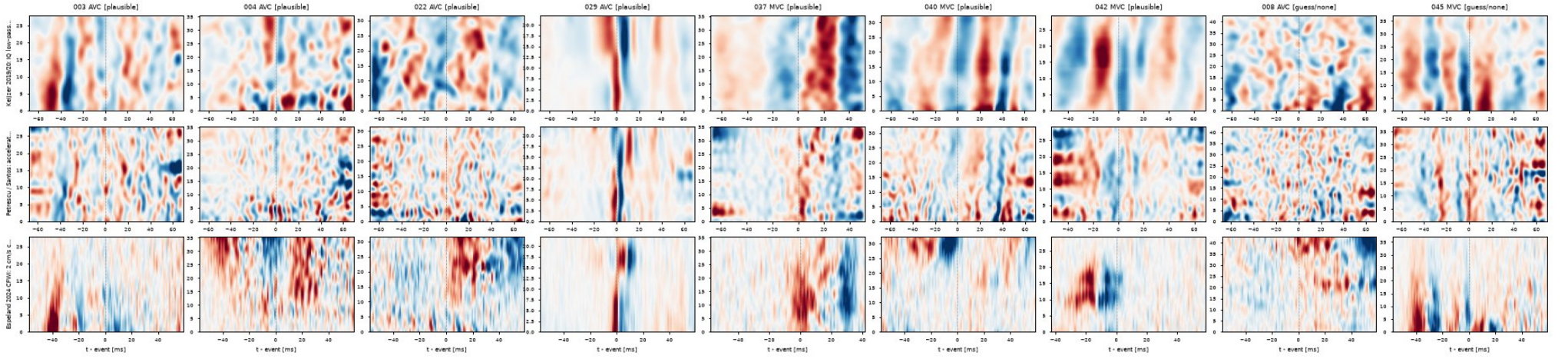




v1 – literature recipes

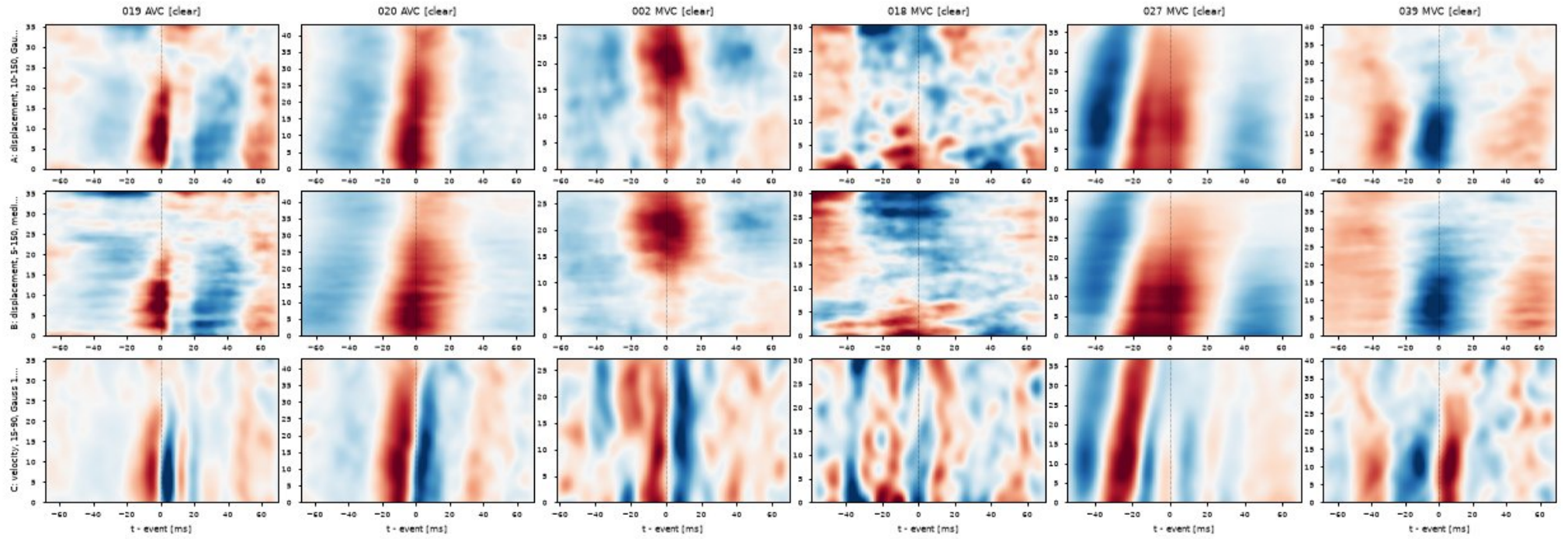
v1 / literature - clear windows - b3 M-lines

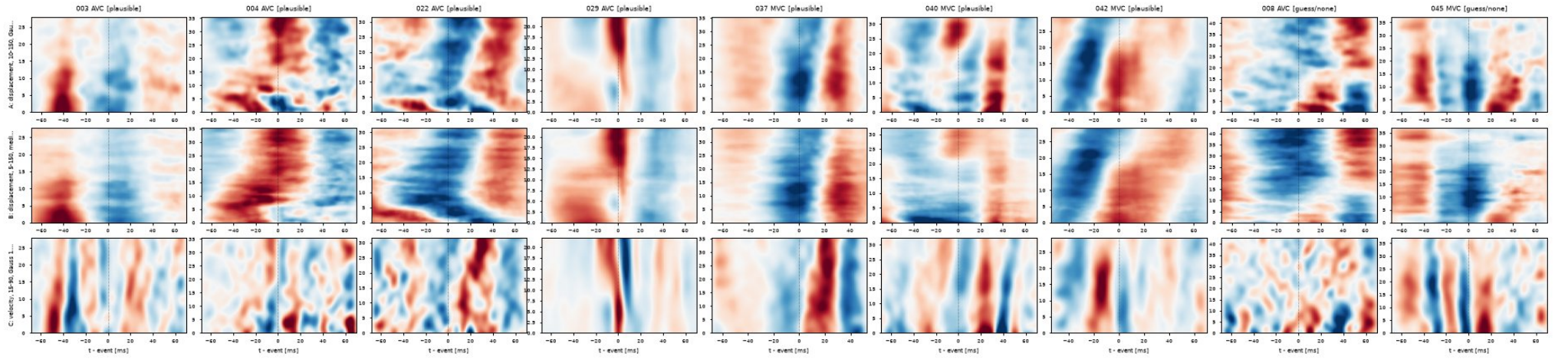




v1 – production views A, B, C

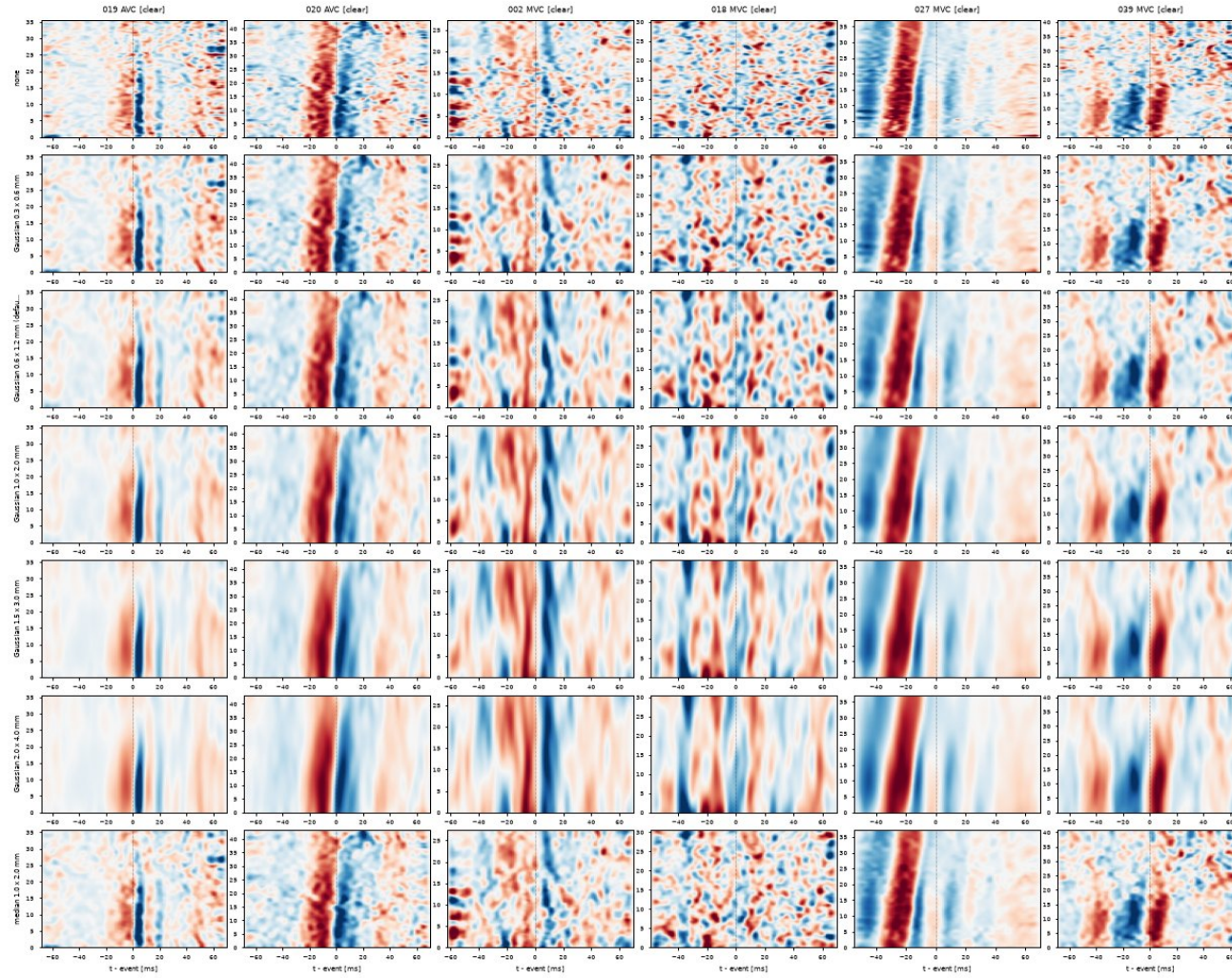
v1 / views - clear windows - b3 M-lines

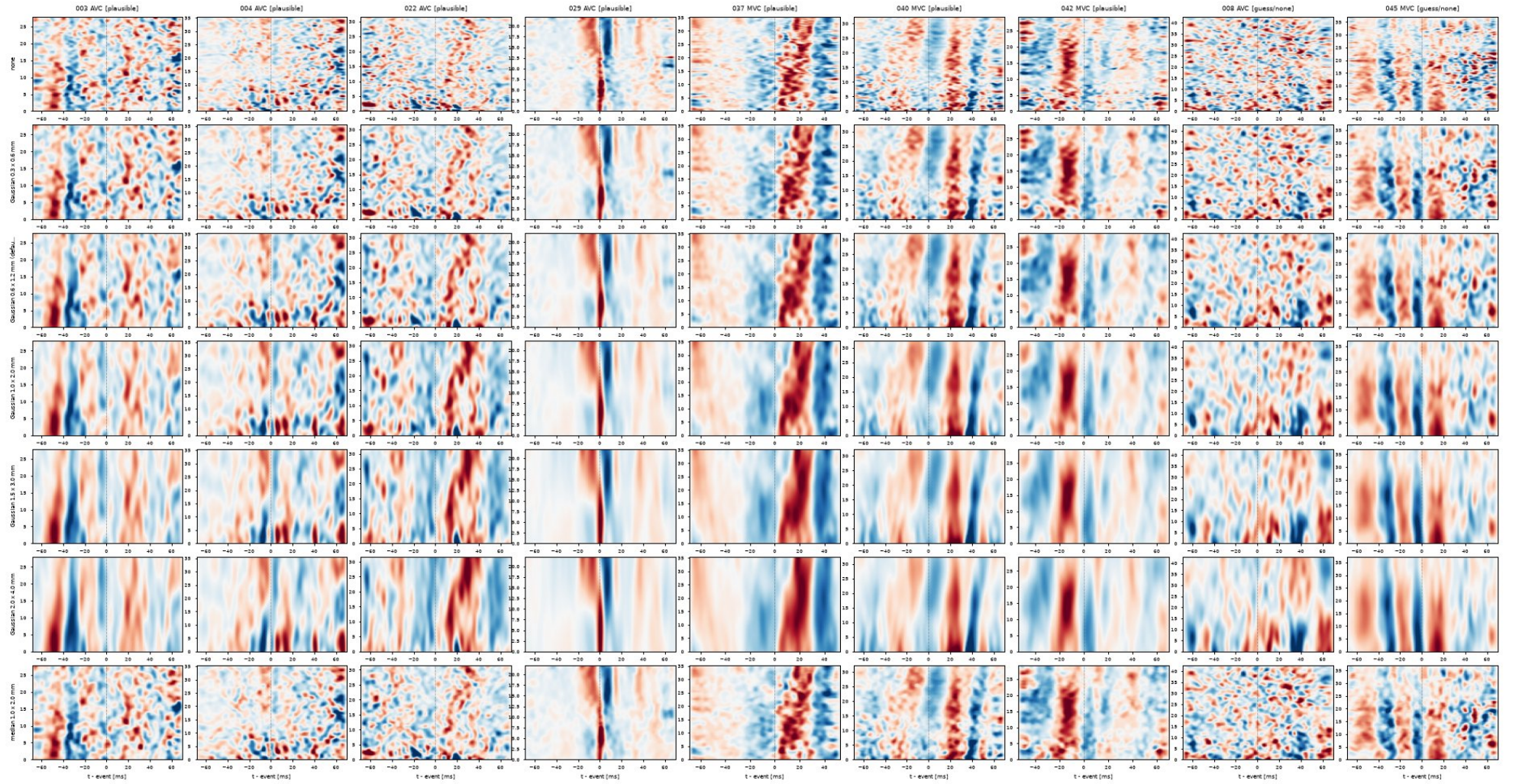




v2 – spatial smoothing (velocity 15–150)

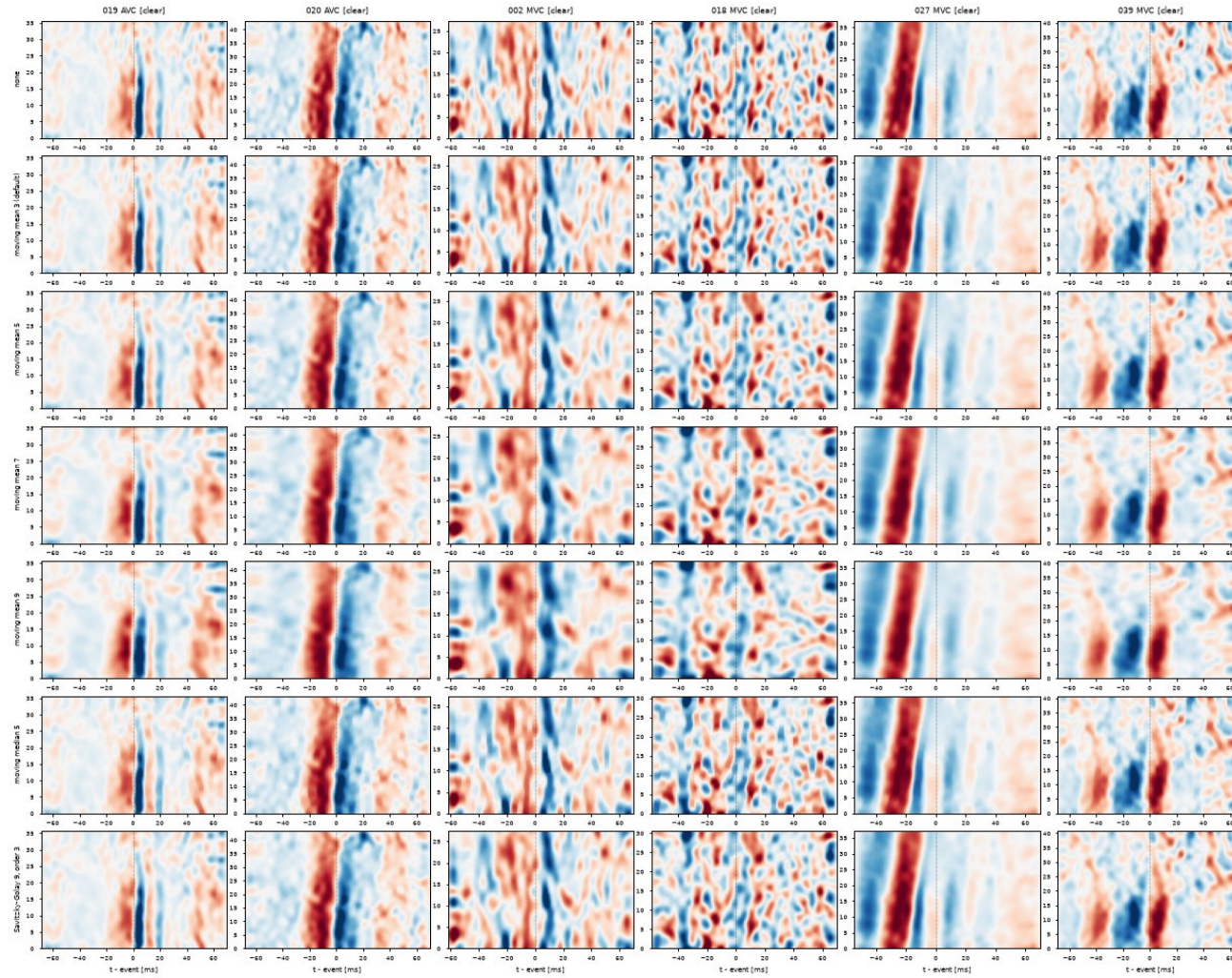
v2 / spatial - clear windows - b3 M-lines

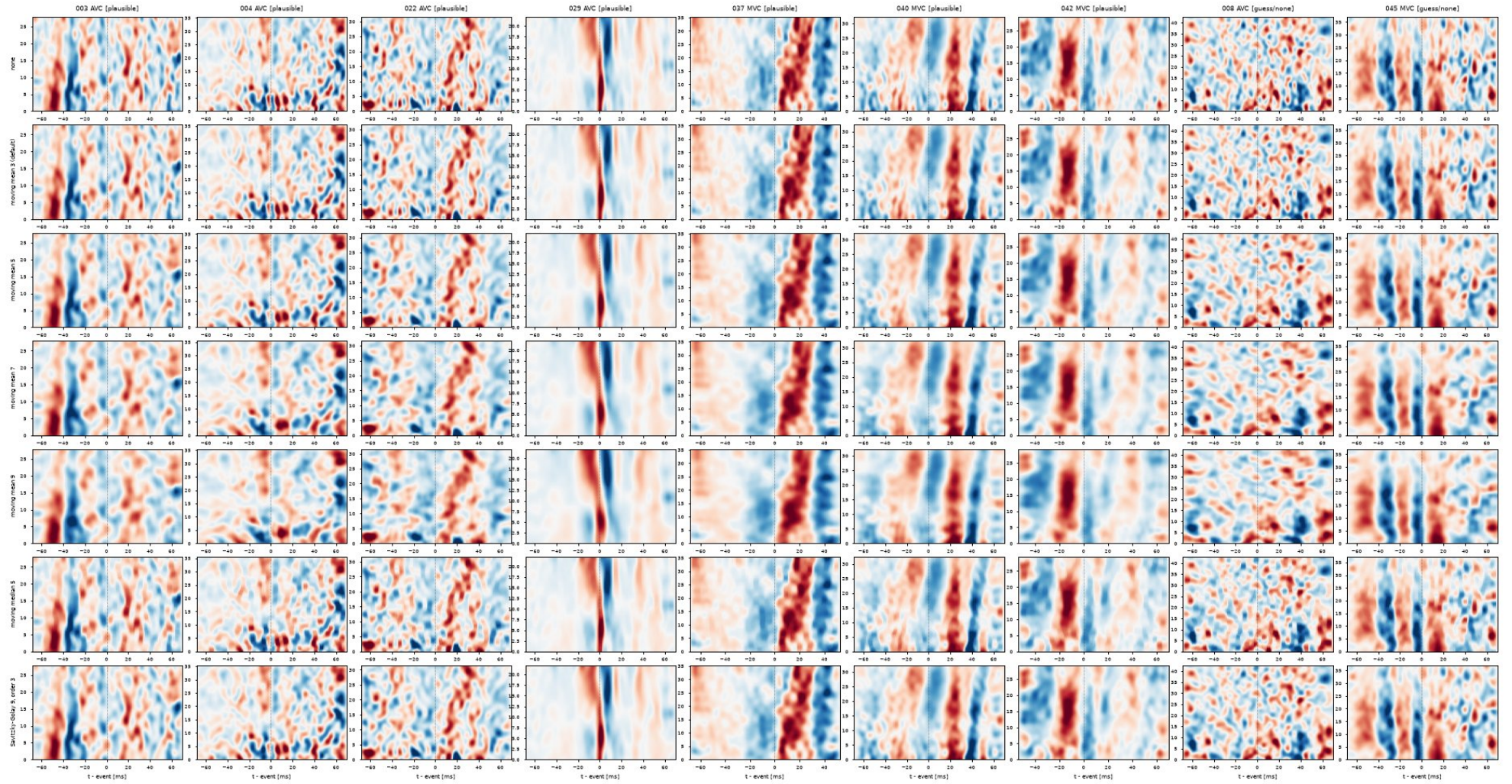




v2 – temporal smoothing

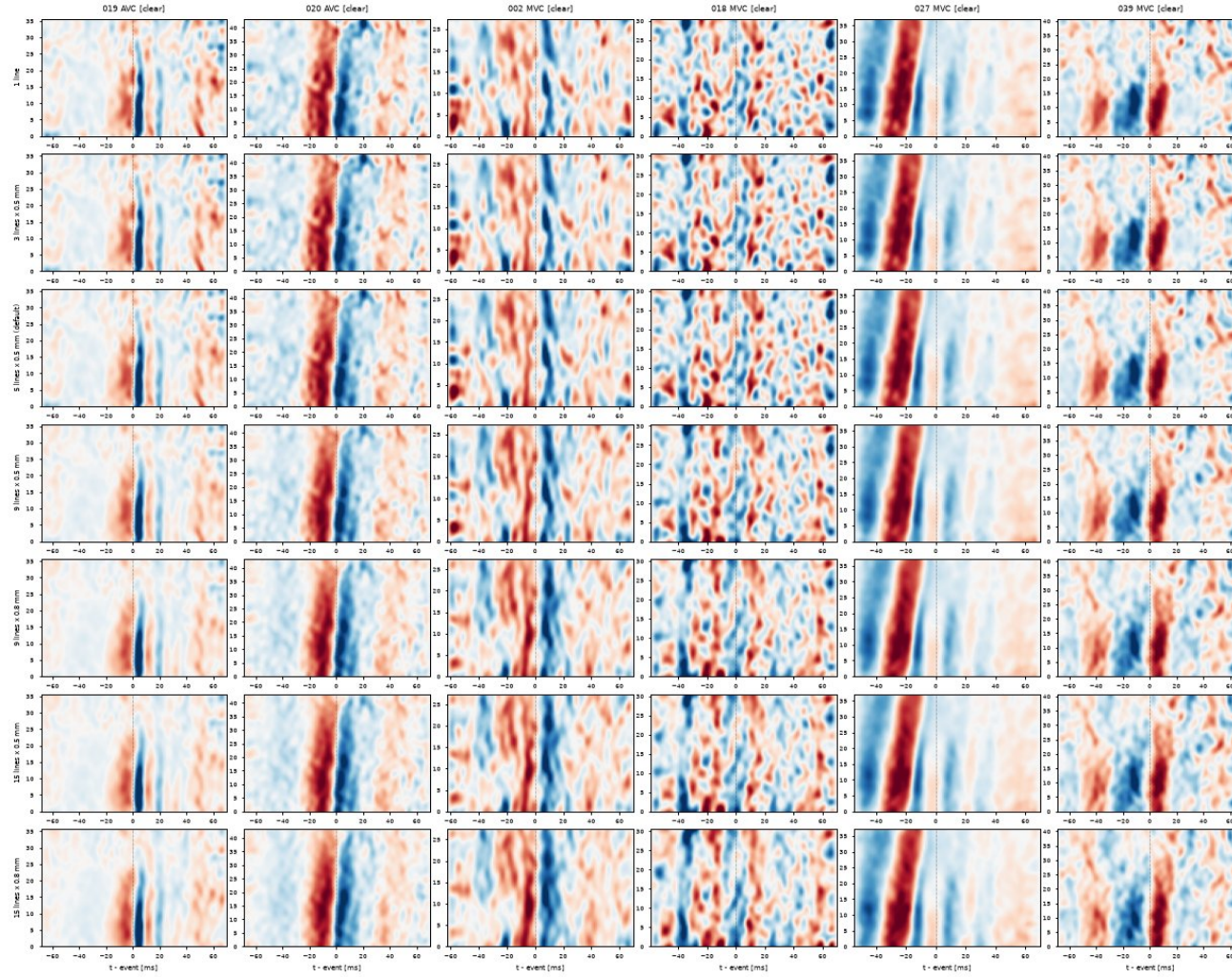
v2 / temporal - clear windows - b3 M-lines

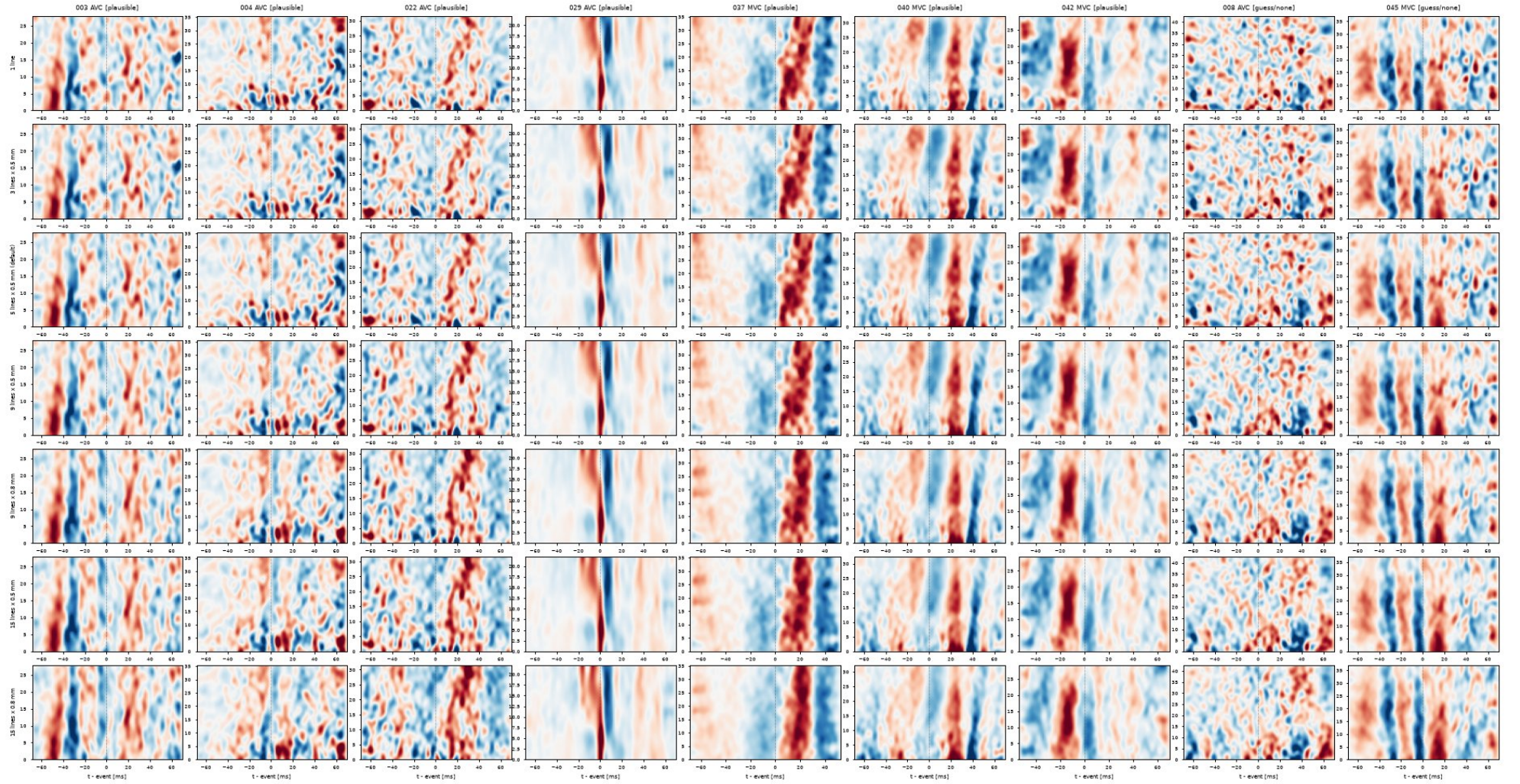




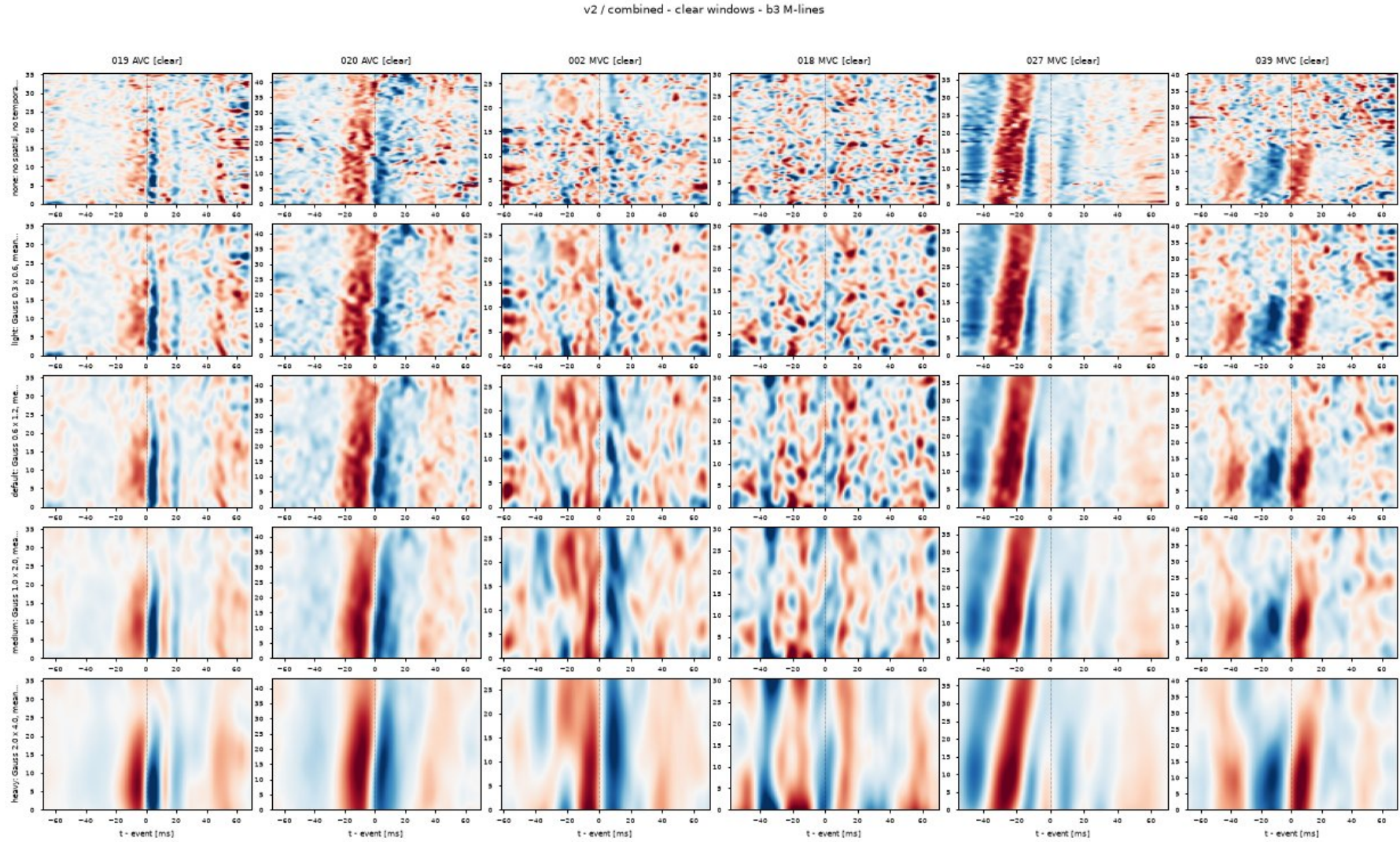
v2 – M-line averaging

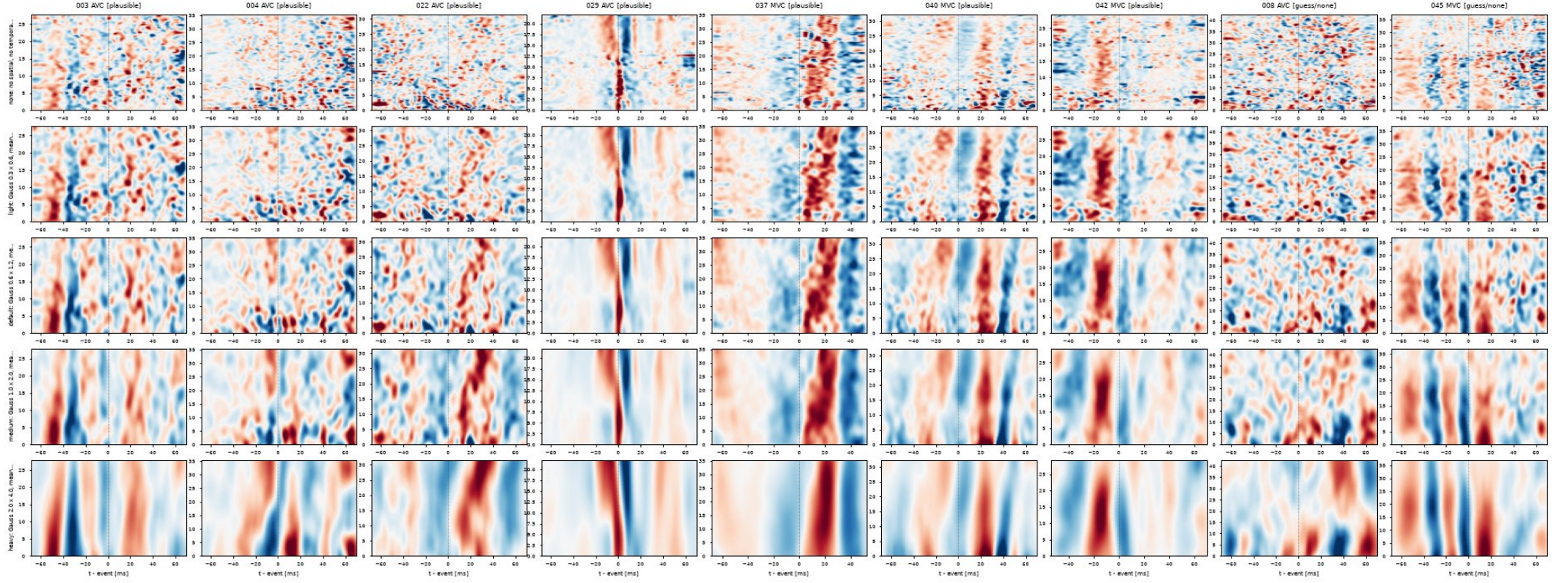
v2 / mline - clear windows - b3 M-lines



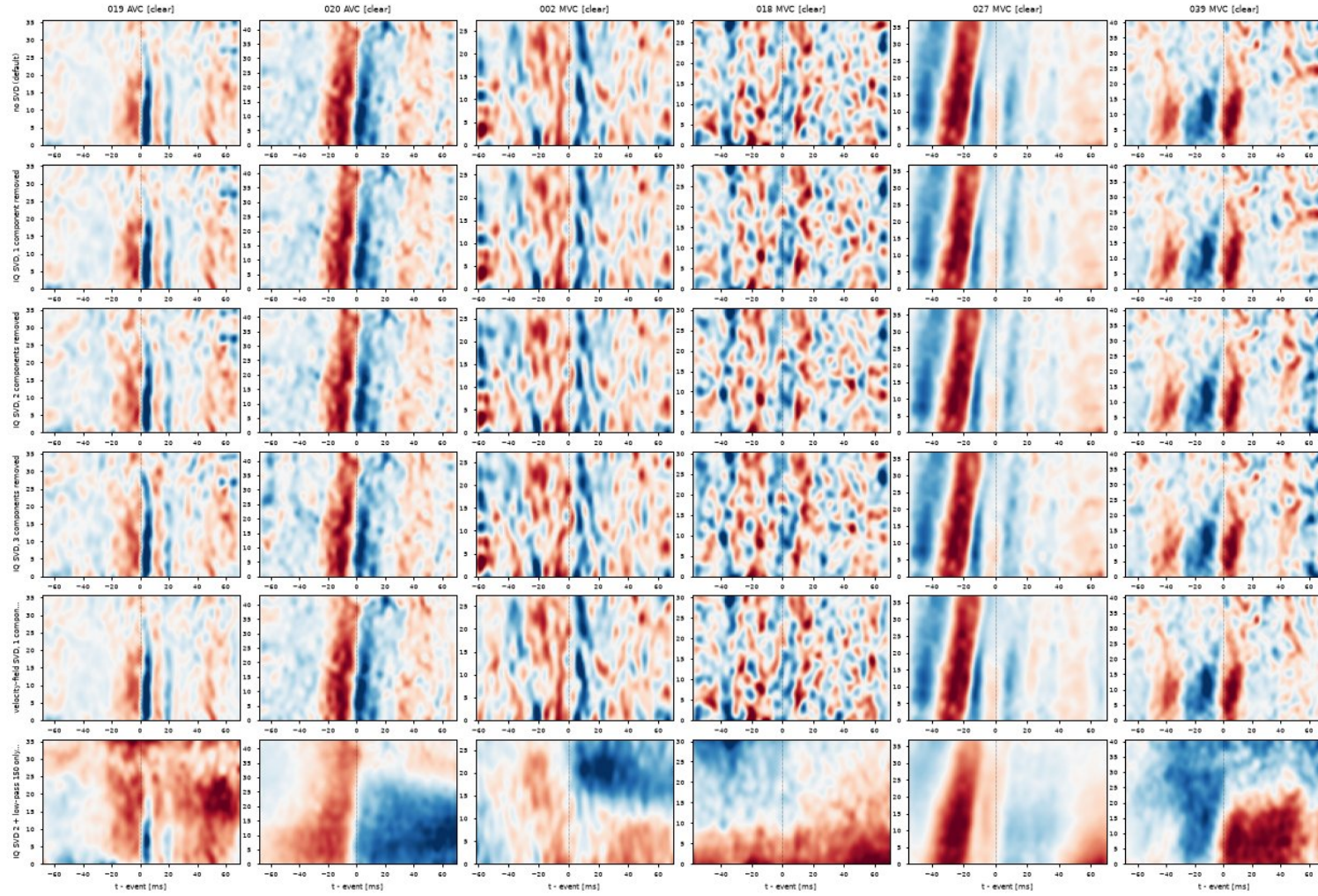


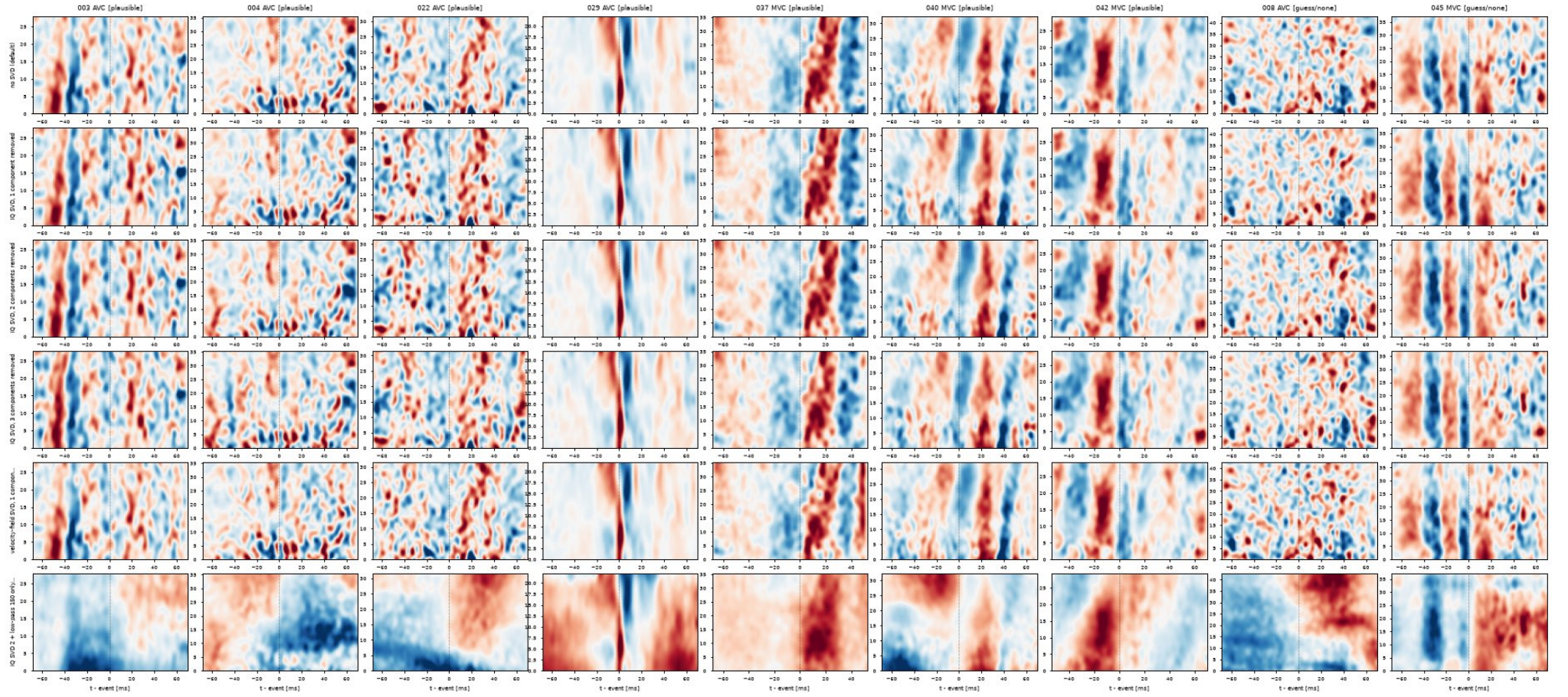
v2 – spatial, temporal and M-lines together





v2 / svd - clear windows - b3 M-lines





v2 / cfwi - clear windows - b3 M-lines

